Explainable AI does not understand a thing: mechanism recovers wiring, not meaning, on a fully known microprocessor — a ground-truth audit of interpretability

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Abstract

Interpretability methods are judged by their faithfulness, by whether they name the true causes of a system's output. Yet on real neural systems there is no ground truth against which to check that judgement. The VCS makes that ground truth available: a bit-exact, differentiable port of the Atari 2600, never built to be interpretable, in which the cause of any output follows from exact intervention. Scoring 30 interpretability methods and an oracle positive control (31 rows, 257 per-game records), we find causal and intervention methods more faithful than gradient and correlational saliency (**0.68** $[\mathbf{0.59, 0.78}]$ versus **0.29** $[\mathbf{0.20, 0.38}]$), the robust separation falling on the position regime where naive gradients vanish (activation patching **1.000** versus vanilla saliency **0.000**), yet saliency scores higher on a plausibility proxy. Yet faithfulness is not the dividing line. Even if every method were faithful — and we make this concrete by turning on the bilinear sampler so that gradient methods become faithful too — the result is uniform: every method delivers causal/mechanical relations, not semantics. An exact method returns a causal-effect map or a data-flow graph, not the computation: a graph-correct circuit need not reproduce behaviour (sufficiency **0.44** on Breakout), and a feature matching a named variable can be causally inert (matched fraction **1.0**, faithfulness **0.04** on Pong). The gap is universal: meaning is not recoverable from the mechanism at all. IEEE 610.12 defines software as code, documentation, and data. Interpretability recovers code-level mechanism; the documentation and data (the semantics) are a separate object, unreachable from mechanism alone. We name this measured residual the *semantic gap*; we measure it, and do not close it. Our semantic labels are imported and only causally verified, and plausibility is a documented proxy. As such, the machine offers a reusable ground-truth benchmark for explainable AI, mechanistic interpretability, AI safety, and software verification. Explainable AI does not understand a thing.

Keywords: interpretability, explainable AI, mechanistic interpretability, faithfulness, ground truth, causal attribution, semantic gap, Atari VCS

1 Introduction

Interpretability now carries real weight, because neural networks drive cars, read scans, and file loans, and before we trust such a decision we ask why. Yet on the systems we most want to explain we cannot check whether the answer is correct: a saliency map for a deep network may look *convincing* and shifts with the action, but this does not mean it delivers *a correct explanation* [1]. So far, the field of explainable AI has converged on *faithfulness*, whether an explanation names a system's *true* causes, as its quality criterion. The difficulty, however, is that on a real neural network there is nothing to measure it against.

Similar short comings were also identified in the field of neuroscience. Jonas and Kording made this vivid by turning the neuroscientist's toolkit on a microprocessor whose every transistor is known [2]. Lesions, tuning curves, and connectomics produced rich structure, and none of it recovered how the chip computes. Without ground truth a method that finds structure cannot be told apart from one that finds the *right* structure, and the same blind spot recurs across explainable AI, where attribution maps are validated against each other or against human intuition rather than against the system's mechanism [3]. Faithfulness has stood in for understanding without anyone being able to test the proxy.

Maier et al. recently built the missing reference, a bit-exact, differentiable port of the Atari 2600 Video Computer System (VCS) that reproduces a real 1977 machine to the byte and pixel [4, 5]. The VCS is software with a known specification, so the true cause of any output is computable by exact intervention. This *intervention oracle* (cf. Section 3.2) returns, for any candidate cause of any output, the true causal effect, measured rather than estimated, over the platform Fig. 1 lays out. On this machine an interpretability claim can be scored right or wrong.

Across the neuroscience, attribution, and mechanistic-interpretability traditions — 30 interpretability methods plus the oracle positive control, 31 experiments investigating 257 per-game records — we place each method on faithfulness against the oracle and on a documented plausibility proxy for how convincing it looks. The numbers confirm the causal view: causal and intervention methods are faithful, gradient and correlational saliency methods are not. The contrast is sharpest on discrete sprite-position outputs, where the naive gradient is provably zero because position is set by strobe timing, not by a value it can flow through. In that position regime activation patching reaches faithfulness 1.000 while vanilla saliency collapses to 0.000 on all six core games, and causal and intervention methods average 0.412 against 0.068 for the gradient and correlational family, a gap of 0.344. The saliency method nonetheless carries the higher plausibility (0.9 versus 0.5), looking most convincing where it is least correct, the region Fig. 4 marks as conviction without correctness.

Faithfulness is not the dividing line. With the same oracle we ask not only *which* methods are faithful but *what* a faithful method delivers: a true causal-effect map or data-flow graph, not the variable's meaning and not the routine's algorithm. To make this precise we call an explanation right only if it is **F**aithful (its claimed causes are the true ones), **S**ufficient (it regenerates behavior under held-out interventions), and **M**inimal at the algorithmic level (right = $\mathbf{F}\wedge\mathbf{S}\wedge\mathbf{M}$, Section 3.4). The committed data show the gap in both directions. An automatically discovered circuit on Breakout is graph-correct and minimal yet does not reproduce the behavior it explains ($\mathbf{F} = 1.0$, $\mathbf{M} = 1.0$, $\mathbf{S} = 0.44$ on Breakout): a correct wiring diagram is not the computed function. A learned feature dictionary on Pong matches a named game variable perfectly (matched fraction 1.0) while being causally almost inert ($\mathbf{F} = 0.04$ on Pong): looking like a known variable is not being how the program computes it. A concept can also be linearly decodable yet provably unused, since what is *present* need not be *used*. On this machine the residual between a faithful attribution and an account of the computation is not philosophy but a number, which we name the *semantic gap*.

A 1977 chip may seem the wrong place to judge methods built for learned networks. We read the VCS as a *necessary-condition screen*, not a predictor of success. Failure here, with perfect ground truth and no learning obscuring the mechanism, is strong negative evidence for the claim that the same method reliably recovers causal mechanisms in harder, less observable systems. The VCS still exhibits the properties that make interpretation hard, from aliased RAM cells to race-the-beam timing; what it does not capture we leave to later work on learned agents.

Because the central claim is sharp, we are deliberate about what we do not claim. We do *not* claim that any method recovered the program's meaning, algorithm, or design from scratch. Every semantic label we use is imported from external annotation sets [6, 7] and only *causally verified* on our substrate, never *discovered* by a method under test, so an alignment method that reports perfect accuracy is right by construction, checking a variable we supplied. Probing shows that information is present, not that it is used [8]; our oracle is the strictly stronger test. The plausibility axis is a documented proxy, not a measured human study. As such, this paper *defines and measures* the semantic gap; *closing* it, recovering a system's documentation and design from the system alone, is companion work, not this paper.

The study thus offers a reusable ground-truth benchmark for explainable AI, mechanistic interpretability, AI safety, and software verification, and it reframes the open problem. Faithfulness is not the dividing line; the universal gap between mechanism and meaning is what remains once faithfulness is secured, and we believe that naming and measuring that residual is a first step toward closing it.

2 Related work

Interpretability is studied in three communities that rarely cite one another, yet all ask what, inside a complex system, caused this output. The neuroscience tradition records and perturbs a system, through tuning curves, lesions, connectomics, and directed influence. Jonas and Kording [2] turned this battery on a microprocessor running classic games and found that it produces rich, publishable structure while missing how the chip computes. Their study reported what the analyses found, not how far each fell from the mechanism; we supply that number (Sec. 4).

The attribution tradition asks, for a single decision, which inputs mattered. Gradient and saliency maps [9–12], class-activation methods [13, 14], occlusion [15–17], and surrogate or game-theoretic attributions [18, 19] form the toolkit, applied to Atari agents to visualise what a policy attends [1, 20]. Yet a counterfactual analysis of those same maps found them exploratory, not explanatory [3]: the highlighted regions need not be the causes. On a learned policy that doubt can only be argued, the true cause being unknown; we settle it where it is computable (Sec. 5).

Mechanistic interpretability, the youngest tradition, aims past attribution at the circuit, the subgraph that implements a behaviour. Path patching, causal mediation, and scrubbing intervene on internal activations [21–24]; alignment search and interchange interventions test whether a hypothesised variable is causally realised [25, 26]; sparse dictionaries and probes read features off the state [8, 27–30]. On a real artifact these prove the most faithful (Sec. 6). Yet a recovered circuit, the same oracle shows, is a correct data-flow graph, not the computed function. The sparse-dictionary branch is itself under scrutiny: SAEBench [31] reports that gains on the proxy metrics used to tune sparse autoencoders do not reliably translate into practical interpretability. We probe the same concern from a different side, asking on a known system whether a recovered feature is causally used rather than only decodable (Sec. 6).

Faithfulness is the criterion most traditions agree on: claimed causes should be the true causes. Jacovi and Goldberg [32] make it precise, yet its diagnostics stay indirect when ground truth is absent. Adebayo et al. [33] sharpened this into the sanity check: a saliency method should change when weights or labels are randomised, and several popular methods do not. Yet such checks are only negative: with no reference map none can certify a right method, and the field optimises a proxy it cannot directly score. Our oracle supplies that map and converts faithfulness into a measured quantity: on the discrete sprite-position regime `activation_patching` reaches faithfulness 1.000 while the popular `vanilla_saliency` collapses to 0.000 on all six core games [34], and the leaderboard plots all 31 methods against the same ground truth (Fig. 4) [35].

The concepts we score against are external. AtariARI labels object positions, scores, and lives from commented disassemblies as a probing benchmark [6]; OCAtari adds offset corrections so the coordinates align with the render [7]. These labels are imported, not discovered, and we verify them causally by intervention — set the byte, re-render, confirm the object moves to its predicted place on the bit-exact framebuffer. This is the hinge of our honesty contract: every concept-level (T3) score measures alignment to a supplied label, not a meaning found on its own. Note that a concept can be linearly decodable yet play no causal role, the "present $\neq$ used" trap that control tasks were built to catch [8]; our intervention test is stronger than probing, and that separation is one of the three counted in Fig. 5.

Ours is not the first study to score interpretability against a known answer, yet existing benchmarks build the answer into the system. Yang and Kim's BAM [36] grades attribution maps against images with known relative feature importance, and M4 [37] extends this idea into a faithfulness benchmark spanning metrics, modalities, and models; in both, the feature ground truth is synthetic, installed by the data-generation or training procedure. Tracr [38] compiles programs into transformers with circuits known by construction; InterpBench [39] trains semi-synthetic transformers to a specified circuit. The Mechanistic Interpretability Benchmark [40] pushes furthest in our direction, scoring circuit localization and causal-variable localization against curated answers in neural language models. Yet each of these answers is engineered for the benchmark, whereas our subject is a real, hand-authored artifact whose ground truth is independent of it: never designed to be interpretable, the VCS yields its causal structure to an exhaustive intervention oracle on a bit-exact, differentiable port [4], built on the Atari substrate [41] and the reference emulator [5]. Our novelty is not the Atari environment, which the Arcade Learning Environment and its descendants [41, 42] already supply, and which faster and object-centric variants such as CuLE [43], OCAtari and JAXAtari [7, 44] extend; it is the bit-exact, differentiable intervention oracle that turns a known machine into a ground truth for interpretability. More importantly, the engineered benchmarks validate faithfulness alone, because there the circuit is the meaning by fiat, whereas on a real artifact the meaning is a separate object from the causal structure. Hence we can do what they cannot: separate faithfulness from understanding, and show that a method which perfectly recovers the true wiring has not recovered the semantics (Fig. 5). We measure that residual, the semantic gap, and are explicit that closing it — recovering a program's variables and design from scratch — is a different problem, the subject of companion work. To our knowledge, this is the first cross-tradition audit of these methods on a real hand-authored software artifact with an exact intervention oracle; we claim no unqualified "first ground-truth calibration," since synthetic known-answer benchmarks and Jonas and Kording's microprocessor study [2] precede us.

Our use of "interpretable" is one of several in circulation. We operationalise it as ground-truth recovery: an explanation is interpretable to the degree that its claimed structure matches the system's true causal structure, scored by the oracle. Barbiero et al. [45] instead define it as inference equivariance, a design target for new models, while ours audits a system already known. Read against Marr's levels of analysis [46] and Lazebnik's "fix-the-radio" parable [47], the point that drives this paper holds: a faithful description of a system's wiring is not yet an account of the algorithm it runs, and on a known machine that gap is measured.

3 Methods

Our method rests on a single idea: we study an interpretability question on a machine whose answer we can compute exactly. We describe in turn the substrate that makes the ground truth obtainable, the oracle that delivers it, the tiered semantics we attach to it, and the correctness triad that turns "do we understand?" into a set of numbers. The per-method setups live in the Supplementary Information; Fig. 1 sketches the pipeline.

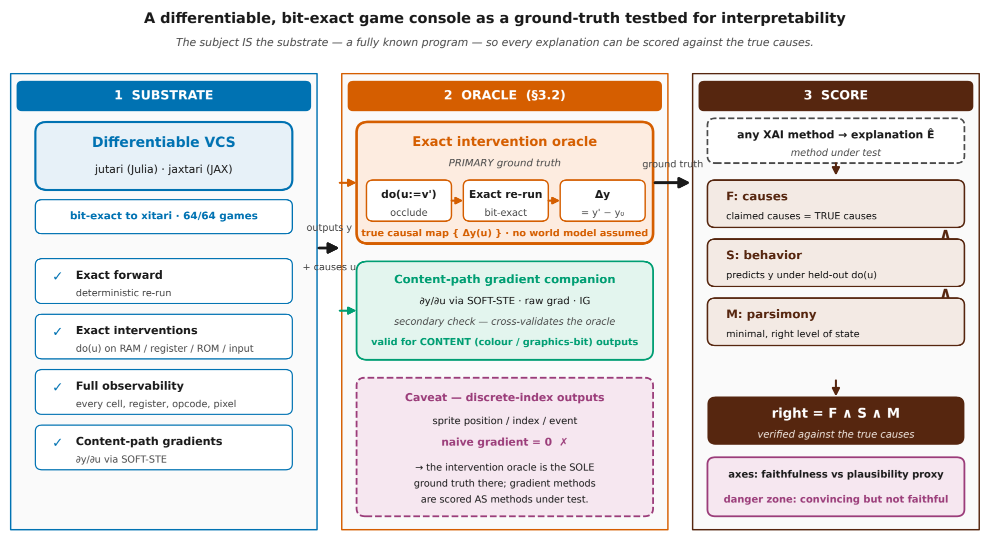


Fig. 1 The measurement platform and its ground-truth oracle. A differentiable, bit-exact port of the Atari VCS (left), realized as `jutari` (Julia) · `jaxtari` (JAX), gives four capabilities no neural network grants for free: exact forward, exact interventions $\mathrm{do}(u)$, full state observability, and content-path gradients. On it we build the oracle (centre): the primary intervention oracle of Section 3.2 (Section 3.2), $\mathrm{do}(u{:=}v') \to \Delta y(u)$ by bit-exact re-run, with a content-path gradient $\partial y/\partial u$ as a companion on the content path only. Any explanation $\hat{E}$ is scored by the correctness triad (right), right $=$ F $\wedge$ S $\wedge$ M, and reported on two axes, faithfulness against the oracle versus a plausibility proxy. Faithfulness aggregates correlation, sign and magnitude agreement, deletion/insertion area-under-curve, and precision at k against the true causal top-k (Section 3.4). A discrete index — a sprite's position, set by strobe timing — has a naive gradient that is zero under the hard forward map (Prop. 1); for position, index, and event outputs the intervention oracle is the *sole* ground truth and gradient methods are scored as methods under test. The schematic draws no experiment data. Data and oracle: see Supplement; oracle defined in Section 3.2 (Section 3.2).

3.1 The differentiable, bit-exact VCS substrate

The subject of this study is the Atari Video Computer System (VCS) itself, not a learned agent: a complex real artifact never designed to be interpretable, yet whose every causal dependency we can read off exactly. Paper 1 built it, a bit-exact, differentiable port of the 1977 VCS validated bit-for-bit against the `xitari` C++ reference on all 64 games of the Paper-1 roster, in random-access memory and rendered screen, within the validated conformance window [4, 5]. We use it in two interchangeable, mutually cross-checked realizations, `jutari` (Julia) and `jaxtari` (JAX/Python).

The substrate grants four capabilities that no neural network offers for free. The first is an exact forward map,

$$y \;=\; \mathrm{VCS}\big(\mathrm{ROM},\, \mathrm{inputs}[0{:}t],\, s_0\big), \tag{1}$$

that is, the output y at step t, a pixel, a score, or a game event, is a deterministic function of the cartridge bytes, the inputs up to t, and the initial state s_0, and re-running reproduces y to the bit. The second is exact interventions: we set any cause u, a ROM byte, a RAM cell, a CPU/TIA/RIOT register, an opcode, or a joystick input, to any value and re-run Eq. (1) deterministically, realizing the causal operator $\mathrm{do}(u)$ on a real machine rather than an assumed world model. The third is full state observability, every bit of RAM and every register recorded along the trajectory, so the "activations" we interpret are the program's own variables. The fourth is a content-path gradient $\partial y/\partial u$ through a straight-through estimator (STE) formulation whose forward pass is bit-exact to the hard machine at any finite temperature [48, 49], batched on the GPU. The inputs we drive, the state we read and clamp, and the outputs we explain are the three faces a neural interpretability study works with, so one substrate carries all three batteries.

We honor one boundary throughout. Paper-1 bit-exactness holds within a bounded frame window on a fixed action stream, on the order of 30 frames of RAM and 60 frames of screen, and every scored claim is taken inside that validated horizon. Long-horizon end-to-end gradients are not needed for our single-step metrics, so we do not assume them.

3.2 The ground-truth oracle

The oracle is the heart of the method: for any output y and any candidate cause u it returns the true effect of u on y, against which every method is then scored. Its primary construction is the exact intervention oracle: we intervene on u by occlusion, clamping, replacement, or resampling, re-run the machine deterministically, and record the change in the output,

$$\Delta y(u) \; = \; \mathrm{VCS}\big(\ldots,\, \mathrm{do}(u{:=}v'),\, \ldots\big) \; - \; y. \tag{2}$$

The true causal effect of a cause is thus the bit-exact difference its surgical edit makes to the output, measured by actually running the altered machine, with no world model or smoothness assumed. The collection $\{\Delta y(u)\}$ over all causes is the true causal map of y, the reference for the faithfulness axis of every method.

The output y comes in five regimes that fix what a method must predict and whether a gradient is even meaningful. A *content* output is a register, color, or graphics-bit value that flows smoothly into the pixel, where the naive gradient is valid. A *position/index* output is a sprite, missile, or ball coordinate set by strobe timing, where the naive gradient vanishes by Prop. 1 and only the bilinear-sampler surrogate or the intervention oracle applies. An *event* output is a discrete state change, a collision latch or a life lost, also gradient-zero. A *score* output is the game's running counter, valid for the gradient only where it varies continuously. A *pixel* output is the rendered framebuffer value, content-valid where it tracks a smooth cause and index-zero where it tracks a position. The Supplement lists, per game, which outputs each method is scored on and the per-output gradient verdict.

The interventions that build the oracle are of distinct manifold kinds, and we state each because the kind bounds what a score means. Occlusion and clamping are surgical but *off-manifold*: they can set a byte to a value the running program would never write. Replacement from a donor trajectory is *on-manifold*: the substituted value is one the machine actually produced elsewhere. Resampling draws the cause from its own marginal over the trajectory, and counterfactual edits are valid under the emulator by construction, since any byte assignment yields a well-defined deterministic re-run. Off-manifold interventions are acceptable for oracle scoring precisely because the oracle measures a *causal* effect, not a likelihood: the question is what the output would be if u were v', and the bit-exact machine answers it exactly for any v', on- or off-manifold. Where a method's own claim is about typical behavior rather than causal dependence we fall back to the on-manifold donor and resampling variants and flag the choice in the Supplement.

The content-path gradient $\partial y/\partial u$, read through the differentiable substrate, is its companion construction. It is fast and dense, and it agrees with the intervention oracle on the content path, the register, color, and graphics-bit values that flow smoothly into the pixel. Yet, a single decisive caveat carries through the whole study. The straight-through gradient routes to the content path only, so a discrete index, a sprite's screen position the VCS sets by the timing of a strobe write, has a naive gradient that vanishes under the hard forward map. We make this precise.

Proposition 1 (Zero naive gradient for discrete-index outputs)**.** *Let an output y depend on a cause u only through a discrete index $n(u) = \lfloor g(u) \rfloor \in \mathbb{Z}$, the integer count of color clocks between a strobe write and the position-counter reset that the VCS uses to place a sprite, missile, or ball, with g continuous, piecewise linear, and non-constant (strictly monotone on each linear piece). Under the hard (bit-exact) forward map the rendered output is $y = h(n(u))$ for a fixed render function h. Then $n(\cdot)$ is locally constant wherever $g(u) \notin \mathbb{Z}$, so $\partial y/\partial u = 0$ at every such point, and y is discontinuous on the set $\{u : g(u) \in \mathbb{Z}\}$, which is measure zero precisely because g is strictly monotone on each piece, and there the gradient is undefined. Hence the naive gradient is zero wherever the derivative exists: an infinitesimal change in u cannot move the sprite by a fractional pixel. Any method that reduces to this naive gradient inherits the failure on position, index, and event outputs.*

The proposition states only the hard-map limit; it does not say that no useful gradient exists. Two surrogate constructions of Paper 1 [4] are explicit exceptions and are *not* covered by it: the straight-through estimator, whose forward pass stays bit-exact to the hard machine at any finite temperature while routing a non-zero gradient along the content path [48, 49], and the bilinear sampler that interpolates the index boundary to recover a usable position gradient [50]. The companion paper constructs both and validates the bit-exactness this proposition assumes; we import that construction and prove the vanishing inline so the present paper stands alone. We therefore replace any blanket "gradients are provably zero" with the scoped claim of Prop. 1: the *naive* gradient for discrete position/index outputs is zero under the hard forward map, and methods that reduce to it inherit that failure, while surrogate and bilinear-sampler gradients are exceptions.

Hence, for content outputs the gradient is a legitimate companion oracle, while for position, index, and event outputs the intervention oracle of Eq. (2) is the sole ground truth and every gradient-based method is scored as a method under test. This position regime is where the sharpest separations in our results appear. Where the two overlap we report their correlation and flag, rather than hide, the points at which they disagree.

3.3 Tiered ground truth and how we obtain it

The ground truth comes in three tiers, and the tiering is not bookkeeping; it carries the central argument of the paper. Tier 1 (T1) is the exact causal map of Eq. (2), derived internally from the substrate and available for all 64 games. Tier 2 (T2) is the hardware roles and the read/write data-flow graph, also exact and internal across all 64 games. T1 and T2 are everything the machine *is* and need no external label.

Tier 3 (T3) is different in kind, and that difference is the point. It attaches a game-concept name to a cell or feature, "the ball's x position" or "the score," names not computable from the substrate alone, because they are facts about what the program *means*, not what it *does*. We import T3 from two public sources and make it causal. AtariARI provides labels for 22 games, read from commented disassemblies, with raw RAM coordinates not aligned to the rendered position [6]; OCAtari covers more than 40 games and, in its RAM mode, applies the render-time offsets that fix that misalignment [7]. We import both as candidates and verify each causally by setting the byte, re-rendering, and confirming on the bit-exact framebuffer that the named object moves to the predicted place, which upgrades a correlational label to a verified-causal one and auto-corrects the offsets; further labels we discover by RAM-to-framebuffer correlation and intervention. The headline core set is six games, Pong, Breakout, Space Invaders, Seaquest, Ms. Pac-Man, and Q*bert, each carrying both sources, all in the Paper-1 bit-exact roster, together spanning paddle-and-ball, shooter-with-shields, resource-meter, multi-agent-pursuit, and discrete-grid logic. The remaining games form a breadth set scored on T1 and T2 only.

We are deliberate about what this buys us. All T3 labels are external and only verified, never discovered from scratch, so a method that matches a supplied concept has aligned to a given meaning, not recovered it. This is exactly why a probe can show that information is *present* without showing it is *used* [8], and why our intervention test, which asks whether the byte *causes* the named object to move, is strictly stronger than decoding it. Recovering the program's documentation and design from the bare machine is a separate problem, the subject of the companion paper.

3.4 The $\mathbf{F} \wedge \mathbf{S} \wedge \mathbf{M}$ correctness triad

A faithful attribution is not yet an understanding, and to make that a measurement we need an operational criterion. We adopt one in the spirit of Marr's levels of analysis [46] and of Jonas and Kording's warning that finding structure is not the same as explaining it [2]: each axis is first a continuous score in $[0, 1]$, and the boolean "right" is a thresholded reading of those scores. We score an explanation $\hat{E}$ of how an output y arises by three numbers, $\mathrm{F}_{\mathrm{score}}(\hat{E})$, $\mathrm{S}_{\mathrm{score}}(\hat{E})$, $\mathrm{M}_{\mathrm{score}}(\hat{E}) \in [0, 1]$, and call it *right* when all three clear their thresholds,

$$\mathrm{right}(\hat{E}) \;=\; \mathbf{1}[\,\mathrm{F}_{\mathrm{score}} \geq \tau_{\mathrm{F}} \;\wedge\; \mathrm{S}_{\mathrm{score}} \geq \tau_{\mathrm{S}} \;\wedge\; \mathrm{M}_{\mathrm{score}} \geq \tau_{\mathrm{M}}\,], \tag{3}$$

with $\tau_F = \tau_S = \tau_M = 0.5$ throughout. Results report the continuous scores; the predicate is a derived summary, not the primary quantity. All three are defined against the oracle, not against any method under test, which keeps the criterion non-circular.

Faithfulness asks whether the claimed causes are the *true* causes, scored against the intervention oracle of Eq. (2). The per-record metric depends on the explanation's output type, so F_{score} is a single number assembled from three type-matched primitives. The committed score is the bare effect-agreement primitive, not a rescaled one: for a real-valued attribution map we report the Pearson (or Spearman, for ranked outputs) correlation $\rho(\hat{E}, \{\Delta y(u)\})$ between the attribution and the true causal effects, scored *raw*, so that a method whose attribution carries no causal signal lands at $\rho = 0$ rather than at the 0.5 of an affine $\frac{1}{2}(1 + \rho)$ remap. This is what makes the position regime informative: under Prop. 1 the naive gradient is identically zero there, its correlation with the oracle is 0, and the reported faithfulness is therefore 0.00 and not a 0.5 floor. The two committed estimators differ only in how the raw correlation is summarized across the six core games. Phase B reports the raw correlation directly (e.g. occlusion at $\rho = 0.482$), so a negative correlation is recorded as a negative number; Phase A's single-unit and tuning analyses, where per-game correlations can be negative, report the per-game correlation clipped at 0 and then averaged across games (e.g. the tuning analysis A3, lowest in Phase A at 0.116, the clip-at-zero mean of six per-game correlations that include negatives). For the same attribution map we also report, as quality measures, deletion/insertion area-under-curve and precision at k against the true causal top-k, $\text{P@}k = |\,\text{top}_k(\hat{E}) \cap \text{top}_k(\{\Delta y(u)\})\,|/k$. For a graph-valued explanation (Phase A connectomics, Phase C circuit discovery) we use the edge F_1 against the true data-flow graph, the harmonic mean of precision and recall over recovered edges, and for Phase C patching and mediation the causal-effect-agreement correlation against the true $\{\Delta y(u)\}$. The final F_{score} is the per-record primary metric averaged first within a game over its output records, then with equal weight across the six core games, so no game-rich method dominates; confidence intervals are 95% bootstrap over games (Section 3.5). The three Phases differ only in which primitive is primary: Phase A uses graph F_1 for the data-flow methods and the clip-at-zero mean correlation for the single-unit and tuning analyses, Phase B uses the raw effect-agreement correlation ρ with deletion/insertion AUC and P@k as the attribution map's quality measures, and Phase C uses the causal-effect-agreement correlation for patching and mediation and graph F_1 for circuit discovery. The leaderboard score is therefore a single scalar per method, but it is multi-metric in origin, and the per-method choice is recorded in the Supplement.

Sufficiency asks whether the explanation could regenerate the behavior under interventions it has not seen. We treat S_{score} as a held-out predictive estimator: $\hat{E}$ is fit (or its causal sites read off) on a calibration set of interventions and then asked to predict the true output y' under a disjoint test set of held-out $\text{do}(u)$ edits, with the prediction compared to the bit-exact re-run. The score is the fraction of held-out interventions whose predicted y' matches the true y' within a fixed output-type tolerance (exact for discrete indices and events, a small ε band for continuous scores), so $S_{\text{score}} \in [0, 1]$ and a value near zero means the explanation predicts no better than the unperturbed output. We deliberately do not define it through interchange interventions or scrubbing, which are themselves mechanistic-interpretability methods under test in Phase C, since that would make the criterion circular; only Phase B and Phase C report S, and its bootstrap intervals follow Section 3.5.

Minimality asks whether the explanation is parsimonious and lives at the algorithmic level of the machine's own module decomposition. We define $\mathrm{M}_{\mathrm{score}} = |U^\star|/|U_{\hat{E}}| \in (0, 1]$, the ratio of the size of the true minimal cause set $U^\star$, the smallest set of registers, RAM cells, opcodes, or program variables whose intervention reproduces y, to the size $|U_{\hat{E}}|$ of the cause set the explanation actually names. An explanation that names exactly the minimal set scores 1; one that records the whole machine state pays for every spurious cell. The target level is the algorithmic unit, the register, RAM cell, opcode, module, or program variable, not the pixel. This is why whole-state recording, which names all 128 RAM cells where the true cause set averages a handful, scores $\mathrm{M}_{\mathrm{score}} = 0.07$ across the six core games: its faithfulness is perfect because it withholds nothing, but it is maximally non-minimal.

Two reporting axes carry the headline comparison, faithfulness against the ground truth and a plausibility proxy. The plausibility proxy is a documented, tradition-level value, not a human-subjects measurement, and we report it as such. It is assigned per method tradition, not per method instance, from the published evidence on how convincing each tradition's explanations are taken to be, before any faithfulness score on our machine is computed, so the proxy cannot be tuned to the result; the per-tradition assignment, its rubric, and its sources are tabulated in the Supplement (Sec. S7), where a sensitivity analysis perturbs the proxy under additive, rank-preserving, rank-jittering, and leave-one-tradition-out schemes and finds the faithfulness-versus-proxy rank correlation stays negative throughout (from -0.79 to -0.40), so the headline contrast survives every alternative proxy assignment we tried. Its job is to expose the danger zone of explanations with high plausibility but low faithfulness, convincing yet wrong. Separating F, which the wiring satisfies, from S and M, which the *computation* must satisfy, is what lets us name, later, the residual that faithfulness alone leaves behind.

3.5 Methods under test, games, and aggregation

We turn three traditions onto the machine and score each by the triad of Eq. (3). The neuroscience battery (Phase A) runs connectomics and data-flow recovery, single-unit lesions, tuning curves, pairwise and global correlation structure, pooled-activity spectra, Granger causality, dimensionality reduction, and whole-state recording, scoring each against the known mechanism to quantify Kording's lesson at the low-faithfulness baseline [2]. The attribution and XAI battery (Phase B) runs gradient saliency and its variants, integrated gradients, occlusion, perturbation masks, randomized masking, LIME, and Shapley-style attribution [9, 10, 15–19]; Grad-CAM and its variants and attention rollout [13, 14, 51] require neural layers or a learned policy and are recorded as not applicable. The mechanistic-interpretability battery (Phase C) runs activation patching and causal mediation, interchange interventions and distributed alignment search, attribution and path patching, automatic circuit discovery, sparse autoencoders, causal scrubbing, linear probing with control tasks, and the logit and tuned lens [8, 21–30, 52].

All faithfulness scoring runs on the six core games of Section 3.3, where the oracle's re-runs are exact. Cell- and pixel-level scores, correlation, deletion and insertion area-under-curve, and precision at k, are taken against T1 and need no T3; only object- and concept-level scores draw on the external T3 labels, and we flag every such score as alignment-to-given, not discovery. We aggregate the committed per-game records into one row per method on the shared faithfulness-versus-plausibility axes of the Results leaderboard. The headline contrast is the position regime, where the naive gradient is zero under the hard forward map by Prop. 1. There, causal and intervention methods average a faithfulness of 0.41 against 0.07 for gradient and correlational methods, a gap of 0.34, while the per-method extreme places an exact causal method, activation patching, at faithfulness 1.000 and the field's default saliency at 0.000 on all six games, the popular method nonetheless carrying the higher plausibility proxy (0.9 versus 0.5). Against these numbers the semantic gap is measured.

4 The neuroscience battery, quantified

We begin where the field's doubt began. Jonas and Kording asked whether a neuroscientist with the standard toolkit could understand a microprocessor, and found that the toolkit yields rich structure that misses the mechanism [2]. They could diagnose the failure but not grade it, because on the 6502 they had no causal ground truth to score each method against. We supply that number. Running the same battery on the bit-exact, fully known VCS, where the true causal role of every register and RAM cell is recoverable by exact intervention (Section 3), turns Kording's qualitative lesson into a measured one. Across the eight analyses A1–A8, faithfulness against the oracle ranges from 0.116 to 1.0 and averages 0.49 (95% bootstrap CI ± 0.24) [35], while the oracle-as-method positive control sits at 1.0 by construction (Fig. 2). The structure these methods report is real; yet structure is not cause, and the gap is what the score measures.

The wiring itself is largely recoverable. A1 reconstructs the dependency graph over state variables, a connectome of read/write edges, and scores it against the true data-flow graph at a faithfulness of 0.41 [35], half spurious. The single-unit lesion A2 fares best: clamping each RAM cell in turn and ranking it by the behavioral damage it does yields an importance map whose Spearman correlation with the unit's true causal role is 0.99 (mean 0.987 over the six core games) [35]. Lesioning is causal, and the one classical number near the ceiling proves it.

Yet a near-perfect rank order hides an interaction blind-spot. A single-unit lesion changes one cell at a time, so it is blind to causes that act only in concert. On Space Invaders the per-unit ranking misses 40% of the interactions the oracle records, and one super-additive pair swings behavior by up to 1044 screen-pixels beyond the sum of its parts [35]. The map of which units matter is right, the model of how they combine absent. Hence even the strongest classical method recovers a ranking, not the computation: an effect map is not the algorithm that produces it.

Tuning curves report structure that points the wrong way. A3 fits per-unit tuning to a game variable, the workhorse of systems neuroscience, and on the VCS it actively misleads. Its *spurious-tuning rate*, the fraction of strongly-tuned units whose tuning does not match their true causal role, is 1.0 on Pong, and its faithfulness collapses to 0.12, the lowest in the battery [35]. Tuning is correlational; on a machine where many cells co-vary with the beam clock, apparent tuning is cheap, which foreshadows the present-versus-used distinction we return to for linear probing.

Correlation structure and field potentials are real and largely epiphenomenal. A4, the spike-word analysis of pairwise and global correlation, reproduces the familiar weak-pairwise/strong-global signature, yet against the true coupling its faithfulness is only 0.29 with sufficiency near zero ($S = 0.018$) [35]. A5, the local-field-potential analogue pooling regional activity into power spectra, scores 0.40, much of that variance being the known frame and scanline clocks [35]. Both see the machine breathe without seeing it compute.

Granger causality mistakes temporal precedence for cause. A6 infers directed CPU/TIA/RIOT edges and scores them against the true data-flow; on Pong its faithfulness is 0.0, with false-edge and missed-edge rates both at 1.0 [35]. The VCS is clock-locked, nearly everything predictable from everything else one cycle earlier, so equating precedence with causation infers spurious edges everywhere. No longer lag or stricter threshold repairs that; precedence and causation come apart here, and only the intervention oracle tells them apart.

Dimensionality reduction lands in the middle, with NMF and PCA tied. A7 decomposes the state tensor into latent components and matches them to the known clock, read/write, and vsync signals; non-negative matrix factorization and PCA both recover a matched-component fraction of 0.60 on average (best 0.80 on a single game), so neither factorization is the more faithful at this analogue [35]. NMF leads only on the secondary recon-error/FSM composite, where its non-negativity better fits the additive register basis. Neither clears two-thirds of the known components, so A7's 0.60, second-strongest behind only the causal lesion, leaves the latent space half interpretable and half opaque.

A perfect record is still not an explanation. A8 records the full RAM and register state, faithful and sufficient by definition ($F = 1.0$, $S = 1.0$), and yet its minimality is 0.07 [35]: a recording of everything is, at the algorithmic level, an explanation of nothing. Faithfulness and sufficiency without minimality is a log file, not an account of the computation, the first hint that the three axes pull apart, which the semantic-gap result later turns into the paper's centre of gravity.

Taken together, the battery confirms the expected result and puts a number on it. The advance over Jonas and Kording is that we grade the *architectural* register-transfer level with a *quantitative* F against an exact oracle, not a qualitative verdict on the transistor. Yet even the battery's best members recover a ranking, a graph, or a record, never the algorithm the program runs. That residual is the universal gap the paper measures. We mean this strictly as calibration: even with perfect ground truth, the classical toolkit finds mechanism, not meaning.

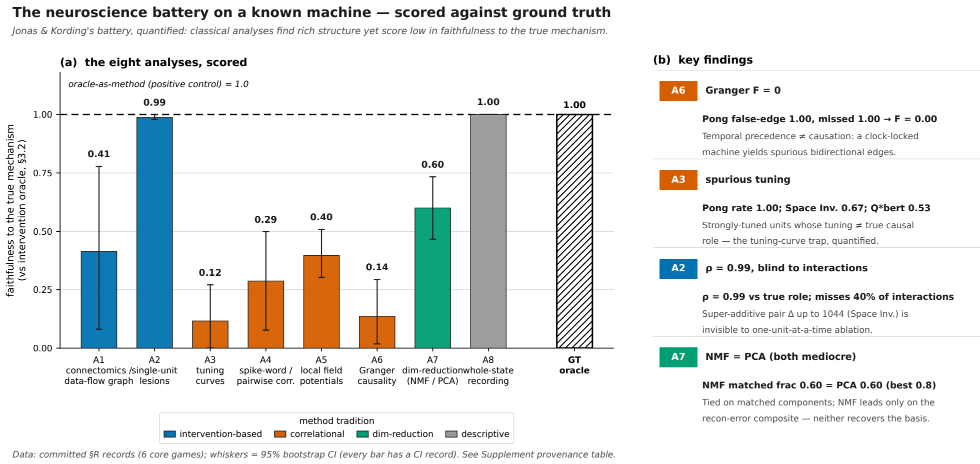


Fig. 2 The Kording neuroscience battery scores well below the intervention oracle. Faithfulness of each classical analysis A1–A8 against the intervention oracle (Section 3.2); the battery averages 0.49, and only the causal single-unit lesion (A2) and dimensionality reduction (A7) sit clearly above chance, while the oracle-as-method positive control reaches 1.0 (dashed reference). Per-metric meanings: A1 = data-flow-graph F_1 vs oracle; A2 = lesion-importance ρ vs true role; A3 = 1−spurious-tuning rate; A4 = coupling correlation vs true coupling; A5 = 1−clock-explained variance; A6 = 1−false-edge rate vs true data-flow; A7 = NMF matched-component fraction; A8 = minimality M. Bars are coloured by method tradition; whiskers are asymmetric 95% bootstrap CIs (`leaderboard_ci.csv`), present on all eight Phase-A bars. Panel (b) is a key-results table isolating the mechanism behind the low scores: A6 Granger causality reaches $F = 0$ on Pong (temporal precedence read as causation on a clock-locked machine); A3 tuning shows a spurious-tuning rate of 1.0 on Pong; A2 attains $\rho = 0.99$ yet misses 40% of interactions, one super-additive pair swinging up to 1044 pixels; and A7 NMF and PCA tie at 0.60 on matched components, both below two-thirds. The figure is the calibration baseline that the causal contrast of Fig. 4 sharpens. Every plotted value reads from a committed per-game record; no number is re-run or estimated.

5 Results B — attributing a VCS output to its causes

Attribution hands back a map of which inputs mattered, and we turn the toolkit on the VCS to ask the one question a real network never lets us ask, whether the map is true. Every attribution is scored against the same intervention oracle (Section 3.2) that defines ground truth, so for each chosen output y, a pixel, a score, a game event, we know the exact set of causes it should have named. We ran twelve attribution methods across six core games (Pong, Breakout, Space Invaders, Seaquest, Ms. Pac-Man, Q*bert) and an oracle-as-method positive control, placed on the shared faithfulness axis in Figure 3 (Phase-B half) and scored per method in Table 1.

The headline is a split, not a ranking. A content output — the value of a register, a colour, a graphics bit — is a differentiable function of the state, so its gradient flows; a position output — where a sprite is drawn, set by strobe timing on a discrete index — has provably zero naive gradient (Paper 1, and Sec. 3). Hence the entire gradient family inherits this boundary, and it cuts the leaderboard cleanly in two.

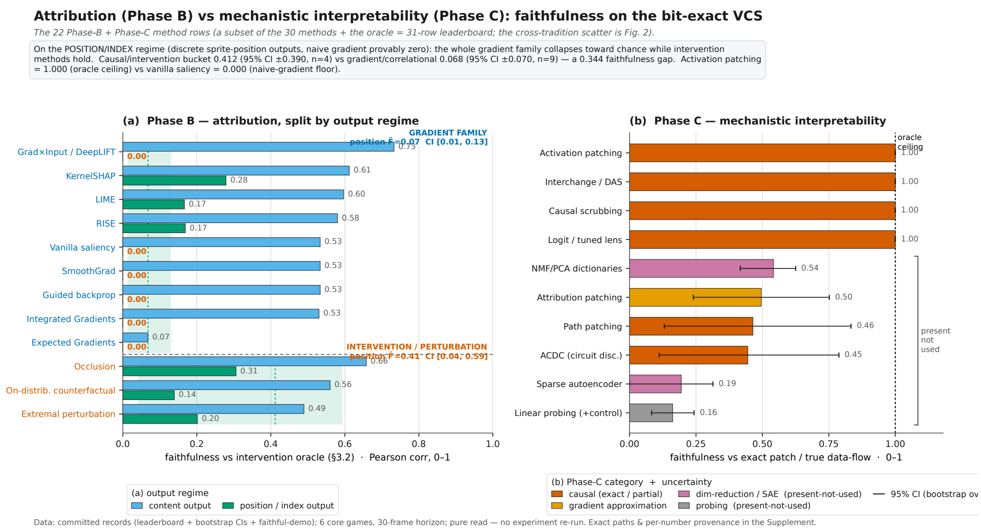


Fig. 3 Attribution (Phase B) and mechanistic interpretability (Phase C) scored against the same intervention oracle (Section 3.2). The Phase-B half (a) splits each method's faithfulness by output regime: on a *content* output the gradient family lands on the true cause, but on a *position*/index output the entire gradient family *collapses to* 0.00, while the intervention methods hold up because each probe is a real re-run of the bit-exact program. Note that the popular gradient methods, which fail on position, carry the higher plausibility proxy, the danger zone of Sec. 7 in numbers. The Phase-C half (b) is discussed in Sec. 6. The figure shows the 22 Phase-B and Phase-C method rows, a subset of the 30 methods plus the oracle positive control (31 rows) of the full leaderboard. Uncertainty bands are 95% bootstrap CIs: per method in Phase C, and at the family level for the Phase-B position-regime means (causal 0.412 [0.04–0.59] vs. gradient 0.068 [0.01–0.13]). Vector figure; values read from the committed cross-tradition leaderboard.

On the content path the gradient is faithful and unremarkable. Vanilla saliency [9] lands its mass on the true causal byte: on Q*bert its attribution of a content read correlates with the oracle at Pearson 0.999 with precision@3 of 1.0, on Ms. Pac-Man at 0.811 / 1.0, on Breakout at 0.750 / 1.0. Integrated Gradients [10], Grad×Input/DeepLIFT [53], SmoothGrad [11], and guided backpropagation [12] all reproduce this, because the output we explain is a one-hot read whose Jacobian $\partial y/\partial u$ is a constant pointing at the read index. Smoothing leaves the score put ($\Delta\,\text{corr} \approx 0$) and guided backprop degenerates to vanilla saliency; on this path there is nothing for the methods to disagree about.

The position output is where the family fails, and it fails by vanishing. The naive gradient is identically zero on every game, so saliency, Grad×Input, SmoothGrad, guided backprop, and Integrated Gradients all collapse to a faithfulness of 0.0 in the position regime — the "Fail / zero on index outputs" prediction of Sec. 3, now a measured number. Averaged over all output regimes the cross-phase gradient family scores only 0.294 faithfulness against the oracle (Fig. 3; Phase-B gradient methods alone 0.298); on the position regime alone, 0.068. Returning the empty map for half a game's outputs is not a partial success but a structural blind spot, exactly where the discrete game logic lives.

Two finer results sharpen the warning. The baseline choice, the perennial worry about Integrated Gradients, is a magnitude effect here, not a faithfulness one: the $(x - x_0)$ factor moves max $|\mathrm{IG}|$, yet the ranking and the oracle correlation hold, because the constant one-hot Jacobian keeps the mass on the same byte; only the degenerate baseline equal to the content byte breaks it. Expected Gradients [54], averaging IG over a distribution of recorded VCS states, removes that degeneracy and is provably stable across reference draws (mean pairwise cosine ≈ 1). Yet, on the content path it scores the same faithfulness as a well-chosen single-baseline IG at $M\times$ the compute and still vanishes on position; the more careful method buys stability, not truth, and its all-regime faithfulness of 0.034 is the lowest in the study.

Guided backprop and vanilla saliency fail the program-randomization sanity check of Adebayo et al. [33] on the system itself. We scramble the ROM so the program boots to a completely different state ($\sim$69–127 of 128 RAM bytes change, lowest on Seaquest), and the content-path saliency map stays byte-identical (cosine ≈ 1.0 on all six games). The Jacobian $\partial y / \partial u = e_{\mathrm{idx}}$ is a constant one-hot, independent of what the program computes, so the map points at the read index no matter which program produced the output, i.e. a saliency that is provably model-invariant. A faithful explanation must change when the model changes; this one cannot, whereas the intervention oracle distinguishes the two programs at once. The map lands on the right cause yet stays blind to the model that produced it.

Replace the gradient with an actual intervention and the position blind spot disappears. Occlusion [15] sets a candidate cause and re-runs the bit-exact program, the coarse twin of the oracle; on the Pong position output, where every gradient method scores 0.0, it recovers the true causal bytes at correlation 0.837, its all-regime faithfulness of 0.482 the highest in Phase B. The black-box perturb-and-re-run methods follow the same logic and result: RISE [17] (0.375), LIME [18] (0.382), and KernelSHAP / Shapley sampling [19, 55] (0.446) each draw masked coalitions and average over the genuine responses, so each works on position outputs where the gradient is dead, and each reports its own completeness ($\Sigma\phi = y(\text{intact}) - y(\text{all-masked})$ for KernelSHAP, the surrogate R^2 for LIME).

The two minimal-mask methods meet Atrey et al.'s [3] objection that perturbations stray off-manifold. Extremal perturbation [16, 56] optimises an area-bounded mask of real *do*(occlude) re-runs, and the on-distribution counterfactual [57] substitutes candidate cells toward a genuine alternative VCS state, the RAM of another frame of the same ROM. Both stay on the data manifold by construction and both work on position outputs (position-regime faithfulness 0.202 and 0.139, all-regime 0.346 and 0.350). As such, the cross-phase intervention family averages 0.681 faithfulness [95% CI 0.59–0.78, $n = 11$] against the oracle, more than double the cross-phase gradient family's 0.294 [0.20–0.38, $n = 14$] (Fig. 3); restricted to the three Phase-B intervention methods alone the mean is 0.393 against the Phase-B gradient family's 0.298. A perturbation is faithful exactly when it is a valid intervention.

A part of the popular toolkit returns no answer at all, because it has nothing on the VCS to attach to. Grad-CAM and Grad-CAM++ [13, 14] need a final convolutional layer to weight; attention rollout [51] needs attention matrices; VIPER [58] needs a learned policy to distil. The VCS is a 6502 CPU with a TIA and a RIOT executing a fixed ROM, so across all six games the count of each required substrate is exactly zero, against ≥ 16 genuine candidate causes per game for the applicable methods. We record this as a measured finding, `applies` = `false`, not a poor score: the absence is structural and the same in every game, and re-targeting Grad-CAM's pooling onto raw register state would only be the content-path gradient renamed. That a slice of the field's headline methods is silent on a real, deployed artifact is itself part of the audit.

Across the twelve methods the leaderboard separates along the axis that matters and crosses the axis the field actually optimizes. The starkest pair makes it concrete: on the position regime the causal method activation patching (Phase C) reaches faithfulness 1.000, the oracle ceiling, on all six games, while vanilla saliency, a widely used baseline, collapses to 0.000 on all six, a per-method gap of 1.000, bit-exact on every game. Yet, on the plausibility proxy of Sec. 3 the popular method carries the higher score (0.9 versus 0.5): the provably empty map looks more convincing than the one that is exactly right. That inversion of low faithfulness paired with high plausibility is the danger zone, and on the VCS it is not a worry but a measurement.

We claim no more than the faithfulness of these maps. A correct occlusion map names the true causal bytes; it does not tell us what those bytes mean, that a cell is the ball's x or that a routine is the bounce. Object-level semantics enter only through the external T3 labels (AtariARI, OCAtari; Sec. 3), and even there we verify a supplied label by intervention rather than discover it. Whether a faithful attribution amounts to understanding the computation we take up in Sec. 7; here we have established only that faithfulness can be had, and that the most popular way of seeking it does not deliver it.

6 Mechanistic interpretability on a known circuit

Mechanistic interpretability makes the strongest claim in the field: not where a system attends, but how it computes. Its methods read internal activations as a circuit and try to recover that circuit, and on a neural network the recovered circuit can only be checked against itself, because the true wiring is unknown. The VCS removes that limitation. Its data-flow is the program's data-flow, known exactly from the disassembly, and every intermediate value is a RAM cell or a register we can read, clamp, and re-run bit-exactly. We therefore turn ten mechanistic-interpretability methods onto a circuit whose answer is already on the page, and score each against the same §4 oracle: the exact intervention table, the true read/write graph, and the $\mathrm{F} \wedge \mathrm{S} \wedge \mathrm{M}$ triad of §3. This calibrates the toolkit against a *known* circuit inside a real, hand-authored artifact rather than a synthetic or compiled one. Figure 3b places the Phase-C methods on the shared faithfulness axis. The result splits cleanly, and the split is the point of the paper: the causal methods are exact, yet exactness buys the wiring, not the meaning.

Exact recovery of the wiring

The headline is bare. Activation patching recovers the true causal-effect table exactly: across all six core games the recovered per-site effect equals the exact intervention oracle to the last bit (`recovered_equals_exact`: true; maximum absolute deviation 0.0), with site precision and recall both 1.0. Clamping a RAM cell to a donor value and

Table 1 Phase-B attribution scored against the intervention oracle (Section 3.2) (six core games; oracle-as-method positive control = 1.0 on every non-degenerate column). *Faith* is the all-regime faithfulness against ground truth and *Faith (pos)* the position/index regime alone, where the naive gradient is provably zero; *Plaus* is the documented method-tradition plausibility *proxy* (Sec. 3), *not* a human measurement. *Note that* the entire gradient family scores 0.000 on position while the intervention family does not, and that the popular gradient methods carry the *higher* plausibility tag — the danger zone, measured. The two family-mean rows are the *cross-phase* tradition families ($n = 14$ gradient/correlational, $n = 11$ causal/intervention across Phases A–C), not the twelve Phase-B rows above; the Phase-B-only gradient and intervention means are 0.298 and 0.393. Values are read from the committed cross-tradition leaderboard.

Method	Tradition	Faith	Faith (pos)	Plaus (proxy)
Occlusion [15]	intervention	0.482	0.306	0.55
KernelSHAP [19]	gradient/black-box	0.446	0.279	0.90
LIME [18]	gradient/black-box	0.382	0.167	0.90
RISE [17]	gradient/black-box	0.375	0.169	0.90
Grad×Input / DeepLIFT [53]	gradient	0.367	0.000	0.90
On-distribution counterfactual [57]	intervention	0.350	0.139	0.55
Extremal perturbation [56]	intervention	0.346	0.202	0.55
Integrated Gradients [10]	gradient	0.279	0.000	0.90
Guided backprop [12]	gradient	0.267	0.000	0.90
SmoothGrad [11]	gradient	0.267	0.000	0.90
Vanilla saliency [9]	gradient	0.267	0.000	0.90
Expected Gradients [54]	gradient	0.034	0.000	0.90
Cross-phase family — gradient/correlational		0.294	0.068	—
intervention		0.681	0.412	—
Grad-CAM / attention / VIPER [13, 51, 58]	N/A	does not apply (no conv / attention / policy)		

re-running the real ROM is, on this machine, literally the oracle's own operation, so it cannot disagree with it. Causal scrubbing inherits the same strength: the true data-flow hypothesis preserves behaviour under resample-ablation of everything outside it on all six games (preserved performance 1.0), a deliberately wrong hypothesis does not (0.52), and the two are discriminated on every game. Interchange interventions with distributed alignment search (DAS) verify the supplied alignment at interchange accuracy 1.0 against a mis-aligned control at 0.0, and the logit lens reproduces the true intermediate value at fidelity 1.0 on every game. Four of the five causal methods sit at faithfulness 1.0, level with the oracle positive control (Fig. 3b). Where a method's operation *is* an intervention, faithfulness is automatic on a fully observable machine. Yet faithfulness is not the dividing line, and the result below shows that even an exact effect table is not an account of what the program does.

Approximate and search-based recovery

The causal account weakens the moment a method trades exact re-runs for an approximation or a search. Attribution patching, the gradient-linear surrogate for activation patching, correlates with the exact patch at only 0.50 and matches it inside the linear regime in just 0.19 of cases; yet its recovered edges stay precise (edge precision 0.95, recall 0.70) because the linearisation errs in magnitude, not in sign of presence. Path patching, run in the IOI style as a single directed do-operation, recovers the directed edge set of the true routine at mean $F1 = 0.46$: on Q*bert it matches the empty direct circuit perfectly ($F1 = 1.0$), while on Pong it recovers none of the six true edges ($F1 = 0.0$) because each receiver's value is re-derived within the horizon, so no one-step path carries a surviving signal. ACDC, discovering the circuit automatically by resample-ablation edge pruning over a threshold sweep, reaches mean $F1 = 0.45$ (ROC-AUC 0.70). A positive control re-runs the identical edge-restricted procedure under the exhaustive do-set and scores $F1 = 1.0$ on every game, so the deficit is a real measurement of single-shot recovery, not a broken scorer. These are the methods the field reaches for when the exact intervention is too expensive, and on a machine where it is cheap they leave structure on the table: precise about what they find, silent about transitively re-derived dependencies, a recall ceiling rather than a precision failure. The gap to the positive control is the quantified cost of approximation against a circuit that is known.

Three separations from the oracle

Here the paper turns. ACDC on Breakout recovers the data-flow circuit at perfect $F1$ against the true routine, and the recovered graph is minimal at the algorithmic level (F= 1.0, M= 1.0). Yet that same graph, scrubbed and asked to regenerate the behaviour under held-out interventions, preserves only S= 0.44 of it (mean scrub-preserved performance 0.32 across games). A circuit can be the right graph and still not be the computed function. The missing object is not a denser or cleaner graph of the same type, because activation patching has already recovered the effect table exactly, with no tuning budget left to spend: a table of per-site effects is not the bounce algorithm. One method is exact and the other is graph-perfect, and between them the behaviour still escapes. This is the separation a "methods underperform" story can never produce, because nothing here underperformed.

The dictionary methods sharpen the gap from the other side. A sparse autoencoder trained on the recorded RAM trajectory reconstructs the state almost perfectly (fraction of variance explained > 0.99 on Pong) and matches every verified game-variable cell it was asked to (matched fraction 1.0). By the metric the field uses to claim a feature "is" a concept, the SAE has succeeded outright. By the oracle it has not: ablating the matched Pong feature and re-running the real ROM gives causal faithfulness F= 0.041, with S= -0.336. The matched feature, removed, does not change the computation it supposedly names. Across the six games the SAE's mean leaderboard faithfulness is 0.19 against a method-tradition plausibility proxy of 0.75, squarely in the §3 danger zone, and the NMF/PCA dictionary contrast repeats the pattern at lower resolution (matched-component fractions up to 0.75 beside near-zero causal use). Note that this separation runs opposite to the first: the circuit was causal but behaviourally incomplete, the feature is perfectly named but causally inert. The gap runs in both directions at once.

Linear probing closes the triptych with the classic trap. With control tasks (Hewitt & Liang's selectivity), probes decode game-concept cells well above chance (mean probe accuracy 0.89, mean selectivity 0.16), confirming that the information is present in the state. Yet presence is not use. Cross-referenced against the exact intervention oracle, the oracle flags 25 concept cells across the core games that the program never causally uses within the window; of those, 6 are also decodable above chance, so a probe would have nominated them as encoded concepts. Breakout's score cell 84 sits in the larger set: it is present in the state and linearly associated with the displayed score, yet provably inert under intervention within the window. The oracle is strictly stronger than the probe: it asks not whether a value can be read out, but whether changing it changes the output. Probing earns a high plausibility proxy (0.85) on a faithfulness of 0.16, the second-worst entry in the danger zone (Fig. 3b). We state the caveat in the same breath: these labels are imported externally (AtariARI, OCAtari) and only *verified* causally, never *discovered*, so DAS's alignment accuracy of 1.0 holds "by construction" because it confirms a supplied variable. The probe tells us a concept is in the representation; only the intervention tells us the program uses it.

Exactness is not the destination

Phase C is the cleanest demonstration of the paper's claim. The mechanistic toolkit, run against a known real circuit, separates into a causal core that is faithful by construction (activation patching, causal scrubbing, DAS, the logit lens, all at 1.0; phase mean faithfulness 0.63, the highest of the three phases) and a recovery layer that is precise-but-partial (attribution and path patching, ACDC) or present-but-inert (SAEs, dictionaries, probing). The three separations are adjudicated by the oracle, not by our preference, and they appear exactly where the field's success metrics report victory: graph-correct yet behaviour-incomplete (ACDC on Breakout F$= 1.0$/M$= 1.0$, S$= 0.44$), named yet causally inert (SAE on Pong matched 1.0, F$= 0.041$), present yet unused (cell 84). The faithfulness advantage that the causal core holds over the gradient methods is itself regime-dependent, and we state that plainly: the gap is robust only on the position-output regime (0.34, 95% CI $[0.02, 0.50]$, excluding zero), and it collapses when the output is content rather than position or when the oracle-equivalent controls are excluded (0.12 and 0.04, both with confidence intervals crossing zero). The separations above survive that caveat because each is a within-method contradiction on a single game, not a cross-family mean. Yet faithfulness is not the dividing line: the residual, the wiring recovered and the meaning withheld, is the semantic gap, which §7 places on the shared axes and the Discussion names as the field's open problem.

7 Across the traditions — and the boundary the leaderboard cannot cross

We have now run three traditions on one machine, and their scores live on a single axis. Faithfulness asks whether an explanation names the true causes, and on the VCS we answer it exactly: the intervention oracle of Section 3.2 re-runs the bit-exact program for every candidate cause and returns the ground truth no real network affords. In the following we place all of Phases A–C on the shared faithfulness-versus-plausibility plane, read off the leaderboard and its danger zone, and then mark, with the same oracle, the boundary that no method, faithful or not, can cross. Faithfulness is not the dividing line. The variable's meaning and the routine's algorithm is a separate object, and on this machine the distance to it is a number. We name that residual the *semantic gap* and measure its symptoms. The correctness triad we apply throughout

— **F**aithful, **S**ufficient, **M**inimal — is defined formally in Section 3.4; here we only read its scores.

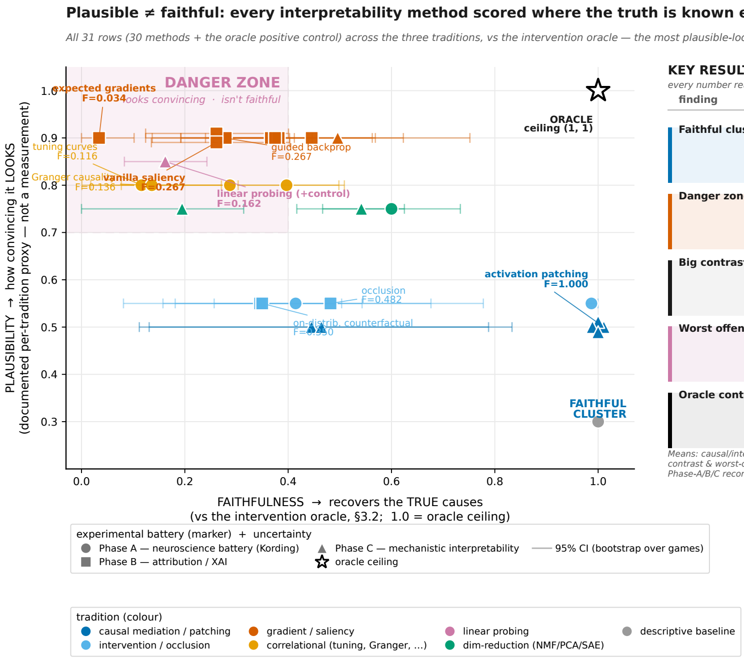
The faithfulness leaderboard

The leaderboard separates cleanly along the axis that matters, and we read it as an exploratory cross-tradition ranking rather than a single commensurable score: the traditions measure different objects, and we report the ranking together with its robustness. Aggregating the 257 committed per-game records into the canonical 30 interpretability methods plus the intervention oracle as a positive control, 31 rows in all [35] (Fig. 4), the causal and intervention methods average a faithfulness of 0.681 against the oracle, while the gradient and correlational methods average 0.294, a gap of 0.387 across all output regimes. On the *position* regime, the discrete sprite-position outputs whose naive gradient is provably zero under the hard forward map (Section 5), the gap holds with the same sign: causal and intervention 0.412 versus gradient and correlational 0.068. The ordering is the one the method matrix predicted (Section 3), now measured. Activation patching, causal scrubbing, interchange interventions, and the logit lens sit at the oracle ceiling of 1.0; the single-unit lesion follows at 0.99; the gradient family and the classical correlational analyses, the tuning curves, Granger causality, and spike-word correlations, fall toward the floor. The starkest pair is bit-exact on every game: on the position regime activation patching reaches faithfulness 1.000, the oracle ceiling, on all six core games, while vanilla saliency, a widely used attribution baseline, collapses to 0.000 on all six, a per-method gap of 1.000 [34]. Hence, on a system where the answer is known exactly, an everyday attribution tool scores near chance where a causal method scores near the ceiling.

The separation survives reweighting, and where it does not we say so. Reading the robustness record [35], the causal-versus-gradient gap stays at 0.387 under both equal weighting by method and equal weighting by game, and its 95% bootstrap interval excludes zero in each case ($[0.32, 0.45]$). Yet two qualifications belong in the same breath. When the oracle-equivalent intervention methods are removed, leaving only coarser approximations, the gap shrinks to 0.123 and its interval includes zero, so the leaderboard's headline contrast is carried in part by methods that approach the oracle operation itself. And the gap is regime-specific: on content outputs, where the naive gradient is valid, it falls to 0.045 with an interval spanning zero, while on the position regime it holds at 0.344 with an interval that excludes it. The conclusion that survives every reweighting is the narrow one: on the discrete position/index outputs, causal and intervention methods recover the true causes and gradient-and-correlational methods do not. This is the cross-tradition analogue of what neural-network benchmarks report when faithfulness is scored against a constructed ground truth [37, 40], with the difference that here the ground truth is an exact intervention on a real software artifact rather than a synthetic or pseudo-label.

The danger zone

The faithfulness axis would be unremarkable if it agreed with how convincing a method looks, and it does not. Plotted against the plausibility proxy of Section 3, the cloud



finding	number	interpretation
Faithful cluster	$\bar{F} = 0.681$	causal / intervention methods recover the true causes
Danger zone	$\bar{F} = 0.294$	gradient / correlational: familiar yet least faithful
Big contrast	0.27 → 1.00	vanilla saliency vs activation patching (default → faithful)
Worst offender	F = 0.034	expected gradients: high plausibility, near-zero F
Oracle control	F = 1.000	intervention oracle anchors the ceiling at (1, 1)

Means: causal/intervention vs gradient/correlational families; contrast & worst-offender numbers cross-checked against the Phase-A/B/C records (see Supplement number-provenance table).

Fig. 4 Every method, every tradition, on the shared faithfulness-versus-plausibility plane. The plotted set is the canonical 30 interpretability methods + the oracle = 31 rows, each placed by its faithfulness against the intervention oracle (Section 3.2; X, exact ground truth) and by its documented method-tradition *plausibility proxy* (Y; Section 3), with a horizontal 95% bootstrap-over-games confidence interval on faithfulness (leaderboard_ci.csv). The plausibility axis is a tradition-level proxy, not a human-subjects measurement. The oracle positive control anchors $(1.0, 1.0)$. Causal and intervention methods cluster at high faithfulness while the gradient and correlational methods cluster low yet *higher* on the proxy — the *danger zone* of high apparent plausibility at low faithfulness. The key findings and per-method values are in the in-figure table; every plotted value reads from the committed leaderboard, none re-run or estimated.

separates into a benign diagonal and a hazardous corner: the methods the field reaches for first sit near the faithfulness floor while carrying the highest proxy tags. Expected Gradients tops the danger list at plausibility 0.9 against faithfulness 0.034; linear probing follows at 0.85 against 0.162; A3 tuning at 0.8 against 0.116. The inversion is the point: the map that is provably empty on the position regime looks *more* convincing than the map that is exactly right. We state the caveat plainly. The plausibility axis is a documented method-tradition proxy assigned per tradition before the faithfulness scores were read, not a human-subjects measurement, and we report it as such to avoid overclaiming the danger zone. Still, the faithfulness axis is exact, and the ranking it inverts is the field's own habit of trust, so the hazard is real even at the proxy's coarse resolution. The danger zone is a measurement, not a worry.

The universal semantic gap

Here the paper turns. Hand the audit its best case, a method that scores 1.0 against the oracle, and ask whether it has given us an understanding of what the program does. It has not, and we are careful about what that claim is and is not. We do *not* claim that no method can recover meaning; that is the unbounded statement, the subject of the companion design-recovery work, which we have not run here. We claim the bounded thing the oracle can prove: even when a method recovers the true causal structure exactly, what it returns is the *wiring* — a causal-effect map, a data-flow graph — and not the *semantics*, the variable's meaning or the routine's algorithm. The gap is universal: it is not a methods-underperformance problem. The missing object is of a different type, not a worse-fit instance of the same one. Following the triad of Section 3.4, we operationalize the gap through three separations, each adjudicated by the oracle: faithful-not-sufficient, named-not-causal, and decodable-not-used. They diagnose distinct semantic failure modes; they are not a complete theory of semantics. The three show the axes pulling apart in *both directions at once* (Fig. 5).

The first separation is graph-correct yet behavior-incomplete: high F and M, low S. Automatic circuit discovery (ACDC) [24] on Breakout recovers a data-flow graph that is, by the exact data-flow oracle, perfectly correct and perfectly parsimonious, $F = 1.0$ and $M = 1.0$ [35]. Yet the discovered circuit alone, with its non-circuit parents scrubbed, reproduces the clean candidate-cell readouts only 44% of the time, $S = 0.44$. A correct graph of which cells feed which is not the function those cells compute. The faithful side shows the same: activation patching [21] recovers the exact per-site causal-effect table on every one of the six games [35], and a table of effects, however exact, is not the bounce algorithm that produced them. No tuning budget is left to spend here, because the effect table is *already* exact, so the residual to the algorithm cannot be a worse fit closed by a better hyperparameter. It is a missing object of a different type, and that is why the gap is not merely "methods underperform."

The second separation is named yet causally inert, the reverse failure: high matched fraction, low causal F. A sparse autoencoder [27] trained on the recorded Pong state matches every one of its candidate game-concept variables, a feature-to-variable matched fraction of 1.0, a perfect semantic alignment by the field's usual yardstick. By the oracle, those same matched features are nearly causally inert on Pong: faithfulness $F = 0.041$ and a *negative* sufficiency $S = -0.336$ [35]. "Looks like the ball's x" is not "is how the program computes with the ball's x." Where the circuit was correct yet incomplete, the dictionary is named yet wrong; the gap runs the other way on the same axis, which a story of methods scoring too low cannot produce. Recent SAE evaluations report the same proxy-versus-use gap on neural language models [31]; the oracle here turns that observation into an exact causal-use test on a known system. We guard against reading semantic success into an aligned variable: the interchange-intervention alignment (DAS) [26] reports an aligned-variable accuracy of 1.0, but that 1.0 is true *by construction*, verifying a *supplied* variable rather than discovering one. A method that confirms a given concept is not a method that found the meaning.

The third separation is decodable yet unused, the present-versus-used trap, where the oracle is strictly stronger than the probe that walks into it: high probe accuracy and selectivity, low intervention effect. Linear probing with control tasks [8] decodes game concepts from the Breakout state at a mean accuracy of 0.72 with positive selectivity over its control; the information is demonstrably *present*. Yet the exact oracle flags a score-related cell that is *present but unused* (cell 84): re-derived within the horizon, its value has no downstream effect on the output [35]. A concept can be linearly readable yet play no part in the computation, exactly Hewitt and Liang's control-task warning; decodability is necessary, never sufficient. Probing answers "is it there," the oracle answers "is it used," and only the second is about how the program works. Linear probing's leaderboard faithfulness of 0.162 paired with a plausibility proxy of 0.85 is its place in the danger zone, and this cell is the mechanism behind that placement.

Read together, the leaderboard plane of Fig. 4 and the failure taxonomy of Fig. 5 make the same point from three sides, and the triad's axes are not redundant: a method can be faithful and minimal yet not sufficient (ACDC on Breakout, $F = M = 1.0$, $S = 0.44$); it can match a named concept perfectly yet be causally inert (the SAE on Pong, matched fraction 1.0, $F = 0.041$); it can flag a concept the program never uses (cell 84, present but unused). The claim does not rest on any single low number, since a lone low S could be a tuning artifact, but on the gap appearing where tuning *cannot* reach — an already-exact effect table — and running in opposite directions on the same axis at once. That two-sidedness is what a "methods are not good enough yet" account cannot generate. We restate the honesty contract here, where the temptation to overread is greatest. We have *not* recovered the program's meaning, algorithm, variables, or design from scratch. Every semantic label we used is external, imported from AtariARI [6] and OCAtari [7]: the substrate only *causally verifies* a supplied label, it does not *discover* the concept, and the provenance record marks each label as imported rather than recovered. Hence, the faithful methods recover the wiring; the meaning is a separate object, and the distance to it is what we have measured. Recovering that meaning is a different problem, and it is the companion paper's, not this one's.

A note on aggregation closes the section, because two committed records report the sparse autoencoder differently and a careful reader will see both. The leaderboard score [35] aggregates over the per-method records, which for the SAE gives faithfulness 0.195 with a 95% bootstrap-over-games interval of $[0.00, 0.31]$ that brackets it; the figures plot this value. The per-game Pong record used in the named-not-causal separation above is a different, finer object, $F = 0.041$ on that one game, and we keep the two distinct: the 0.195 is the leaderboard aggregate, the 0.041 is the single-game illustration. Throughout the paper, the leaderboard and its figures use the per-method aggregation as canonical, so text, table, and figure agree.

8 Discussion

We have turned the interpretability toolkit onto a machine whose ground truth is computable, and that let us measure two things at once. On a system where the true cause of any output is recoverable by exact intervention, causal and intervention methods are faithful while the field's familiar gradient and correlational tools are not, and the leaderboard separates them along the only axis that admits an exact answer (Fig. 4). A faithful causal account scores 1.0 where the field's default scores 0.0, on a machine whose answer is known exactly. Yet faithfulness is not the dividing line. Even a method at the oracle ceiling returns the causal wiring, an effect map and a data-flow graph, and not the semantics, the variable's meaning or the routine's algorithm. The residual is a number, and we name it the *semantic gap*.

Three traditions that rarely share a page, a neuroscience battery, the attribution toolkit, and mechanistic interpretability, occupy a single plane once a common oracle scores them. Jonas and Kording warned that an analyst armed with systems neuroscience could fail to understand even a microprocessor, because finding structure is not the same as explaining it [2]. We make that a measurement: the classical battery recovers rich structure yet scores low against the known mechanism, the gradient family collapses to the floor on the discrete position regime, and only the causal operators reach the ceiling. Mechanistic interpretability has until now been calibrated against synthetic or compiled ground truth, transformers with a planted circuit by construction [38, 39], and ground-truth attribution benchmarks likewise plant the important features [36]. We add the missing case, a real, hand-authored, deployed artifact never designed to be interpretable, whose ground truth is independent of the artifact rather than built into it. Hence the leaderboard calibrates these methods against ground truth on a system the field did not get to plant.

Yet the same oracle marks a boundary the leaderboard cannot cross, and that boundary is the subject of this paper. A perfect faithfulness score certifies that a method named the true causes, not that it recovered what the program means. We saw the boundary three times, from three sides (Fig. 5): on Breakout a circuit that is graph-correct and minimal ($F = M = 1.0$) yet reproduces behavior only 44% of the time ($S = 0.44$); on Pong a feature dictionary that matches every named variable (matched fraction 1.0) while being nearly causally inert ($F = 0.041$, $S = -0.336$); and a concept that is linearly decodable yet, by the oracle, never used in the computation. The gap runs in both directions on the same axis at once, correct-but-incomplete and named-but-wrong, which is precisely what a "methods are not good enough yet" story cannot produce.

IEEE 610.12 and what understanding requires

Faithfulness stops short for a reason that Marr's levels of analysis make clear [46]. A faithful attribution lives at the implementational level, in which cells and registers carry the signal; understanding lives one level up, at the algorithmic level, in what function the program computes. A correct data-flow graph is an implementational fact; the bounce algorithm it implements is an algorithmic one, and the two are objects of a different type. This is Lazebnik's lament, that one can characterize a system in exhaustive mechanistic detail and still not understand it [47]. We turn that lament into a measured quantity: the distance from a faithful effect table to the algorithm is the sufficiency gap $1 - S$, and on an already-exact effect table it cannot be closed by a better hyperparameter.

The bar is sharpest when borrowed from software engineering. IEEE 610.12 defines software as programs plus their documentation, the specification, the design, the intent [66]. Interpretability recovers code-level mechanism. A faithful circuit recovers part of the program; understanding the system means recovering its documentation, what each variable is for and what each routine is meant to do. Our causal oracle cannot reach that object, because the meaning of a variable is not a causal fact about the variable but a fact about the design it serves. The documentation and data (the semantics) are a separate object — unreachable from mechanism alone. Hence the gap is not an artifact of weak methods but a universal property of the question: faithfulness is not the dividing line. We are deliberate that this paper defines and measures the gap. Closing it, recovering the program's documentation and design from the bare machine, is the reverse-engineering bar and the charge of the companion design-recovery paper (Paper 4), which measures the distance to that bar rather than clearing it. Whether a recovered concept matches how the system is used is the companion behavior paper (Paper 3), and the learned agent, where there is no disassembly to check against, is Paper 5.

Probing reads the same result from the data side: it shows that a concept is present in the state, not that it is used, the control-task warning of Hewitt and Liang [8]. Our intervention oracle is strictly stronger, asking whether the byte causes the output, and it found a present-but-unused score cell the program never reads. Decodability is presence; use is an algorithmic fact; understanding needs the second, and only the oracle supplies it. Note that the three traditions are, at bottom, one toolkit under different names, as tuning curves are feature visualization, lesions are ablations, and Granger causality is the correlational ancestor of causal attribution. Each pair shares a failure mode, so a result measured on one transfers as a caution to the others, and a benchmark built on an Atari chip speaks to systems neuroscience, explainable AI, mechanistic interpretability, and software verification at once.

Why a 1977 chip is a fair screen

The obvious objection deserves a hearing. The VCS is deterministic, hand-coded, and small, while the systems these methods target are learned, distributed, stochastic, and large. We do not ask a score here to predict a transformer. The benchmark is a necessary-condition stress test, not a sufficiency predictor: when a method fails here, with perfect ground truth, full observability, exact interventions, and no learning to muddy the picture, that failure is strong negative evidence for any claim that the same method reliably recovers causal mechanisms in harder, less observable systems. Passing here is necessary, not sufficient; failing here is telling. The same logic suggests, rather than proves, a reading of the semantic gap. If even the strongest causal account a fully observable machine permits stops short of meaning, the result suggests that faithfulness alone should not be treated as understanding in less observable neural systems, where the circuit itself is uncertain and no clean ground truth is on hand. We measure $1 - S = 0.56$ on an exactly recovered circuit on Breakout ($1 - 0.4444$; the ACDC family mean is $1 - 0.32 = 0.68$), and we do not extend that number to neural networks as a proven floor; the precise scope, and what this paper does not claim, are set out in the Supplementary box on what we do and do not measure (S6).

The chip also exhibits the properties that make real systems hard to interpret: aliased, polysemantic RAM cells reused across routines, a hand-coded echo of superposition; computation by racing the electron beam, distributed and temporal; registers that mean different things in different contexts. Figure 6 maps each VCS property to the neural-network difficulty it does and does not capture, so that a property provable here is one suspected there. What the chip does not capture we state without hedging: learned features, scale, and stochasticity are absent, and the learned agent that folds all three back in is Paper 5.

What interpretability needs

Three recommendations follow, and we state them flatly. The field should prefer causal over correlational evidence, because the methods it reaches for first carry the highest plausibility proxy and the lowest measured faithfulness, and a map provably empty on the position regime looks more convincing than the map that is exactly right. It should validate on known systems before trusting on unknown ones, since a method that cannot pass where the answer is computable cannot be audited where it is not. Most important, it should stop treating faithfulness as understanding. The open problem for explainable AI is not a better saliency map; it is the semantic gap. Faithfulness is not the dividing line; meaning is the object that mechanism cannot reach.

We have measured a gap, not closed it, and several limits bound what we may conclude. We did not run semantic recovery, as Phase E, documentation and design recovery, is specified but not executed; our claim is therefore the bounded one, that on exact ground truth faithfulness certifies the wiring and the wiring alone is not the semantics, and not the unbounded "no method can recover meaning," which is Paper 4's charge. The sufficiency and minimality axes are defined against the oracle, and the T3 semantic labels are external, imported from AtariARI [6] and OCAtari [7] and only causally verified, never discovered. A single low sufficiency could be a tuning artifact, so the claim rests not on any one number but on the gap appearing where tuning cannot reach and running in both directions at once. The plausibility axis is a documented method-tradition proxy, not a human-subjects measurement, reported at that proxy's coarse resolution. Our scores hold within Paper 1's validated conformance window; outside that horizon the substrate is not certified bit-exact, and we keep every scored claim inside it. The subject is one old machine with no learning, which is the necessary-condition framing once more: a stress test, not a forecast.

We close by reaching past the present result. Because the substrate is differentiable, it tells us which gradient explanations are trustworthy and where they vanish, as the content path carries an honest gradient while the discrete index does not, so a saliency map over sprite position is empty by construction, not by accident. That statement about gradients as a class travels to any system where a discrete decision sits downstream of a continuous one. We believe the value of a fully known, differentiable testbed is exactly this, that it lets the community calibrate its instruments before deploying them where calibration is impossible. The hard case remains the learned agent, where there is no disassembly, no specification, and no engineer's intent to recover, and where the semantic gap can only widen. It is foreseeable that closing it, turning a faithful account of a learned system into an understanding of what it has learned, will be the work of the next decade of interpretability. We have not solved it. We have made it a number, and a number is where understanding begins.

Data availability. The interpretability scores reported here are derived from the committed analysis records under `tools/xai_study/` (the per-method, per-game §R records and the aggregated cross-tradition leaderboard [34, 35]); every quantitative claim in the Article reads from one of these committed artifacts, and no number is re-run or estimated. The game cartridges used as the subject are third-party Atari 2600 ROMs that we do *not* redistribute; following the practice of the underlying emulator port [4], we acquire them through the `AutoROM` retrieval procedure and report a SHA-256 hash table for each ROM in the Supplementary Information, so that the exact binaries can be reproduced and verified independently. The recorded state trajectories used to train the sparse autoencoders and to run the patching and probing experiments are regenerated deterministically from those ROMs and the published action streams.

Code availability. The interpretability benchmark introduced here—the scoring suite, the $\mathrm{F} \wedge \mathrm{S} \wedge \mathrm{M}$ triad metrics, the intervention oracle, and the per-phase method implementations (`tools/xai_study/`)—is available for review at https://github.com/akmaier/UnderstandingVCS, together with the reproducibility appendix that maps every reported number to its committed record and run command. A versioned snapshot will be archived on Zenodo with a minted DOI, and an optional Code Ocean capsule reproduces the leaderboard and the figures from the committed result records alone. The differentiable, bit-exact emulator that the benchmark rests on, in its two interchangeable realizations (`jutari`, Julia; `jaxtari`, JAX/Python), is already public: it is described in the companion port paper [4] (arXiv:2606.22447) and its source is openly available at the same repository. The companion paper supplies the bit-exactness validation and the gradient caveat that the present work imports; what is new here is the interpretability benchmark, the oracle scoring, and the semantic-gap diagnosis. We release the code under a permissive open-source licence so that a third party can score a new method against the same ground-truth oracle.

Author contributions. A.M. conceived the study, designed the ground-truth audit and the $\mathrm{F} \wedge \mathrm{S} \wedge \mathrm{M}$ triad, implemented the benchmark and oracle scoring, and wrote the manuscript. S.B. contributed to the method design and the analysis and revised the manuscript. P.K. contributed to the differentiable substrate and oracle and to the gradient-faithfulness analysis, and helped develop the differentiability framing. All authors discussed the results, interpreted the semantic-gap finding, and approved the final manuscript.

Competing interests. The authors declare no competing interests.

Ethics and broader impact. This study uses no human-subjects data; the plausibility axis is a documented plausibility proxy by method tradition rather than a measurement, and we report it as such. The subject is a fully deterministic, hand-coded program with no learned policy, so no agent is trained. Interpretability methods are dual-use, in that a tool that audits a system can also be used to probe it; our contribution is a calibration that makes explanation claims more, not less, trustworthy, and we release it openly to that end. The game cartridges are not redistributed.

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References

[1] Greydanus, S., Koul, A., Dodge, J. & Fern, A. *Visualizing and understanding Atari agents. Proceedings of the 35th International Conference on Machine Learning (ICML)*, Vol. 80, 1792–1801 (PMLR, 2018).

[2] Jonas, E. & Kording, K. P. Could a neuroscientist understand a microprocessor? *PLOS Computational Biology* **13**, e1005268 (2017).

[3] Atrey, A., Clary, K. & Jensen, D. *Exploratory not explanatory: Counterfactual analysis of saliency maps for deep reinforcement learning. International Conference on Learning Representations (ICLR)* (2020).

[4] Maier, A., Bayer, S. & Krauss, P. A differentiable Atari VCS: A complex, fully known ground truth for explainable AI. arXiv:2606.22447; code at https://github.com/akmaier/UnderstandingVCS (2026). Companion port paper (Paper 1) of this program; submitted 2026-06-21. Code: https://github.com/akmaier/UnderstandingVCS. Citation key `maier2025vcs` retained for stability., 2606.22447.

[5] DeepMind. Xitari: An Arcade Learning Environment fork. https://github.com/google-deepmind/xitari (2016). Accessed: 2026-06-16.

[6] Anand, A. *et al. Unsupervised state representation learning in atari. Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 32 (2019).

[7] Delfosse, Q., Blüml, J., Gregori, B., Sztwiertnia, S. & Kersting, K. OCAtari: Object-centric Atari 2600 reinforcement learning environments (2023). 2306.08649.

[8] Hewitt, J. & Liang, P. *Designing and interpreting probes with control tasks. Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2733–2743 (2019).

[9] Simonyan, K., Vedaldi, A. & Zisserman, A. *Deep inside convolutional networks: Visualising image classification models and saliency maps. International Conference on Learning Representations (ICLR) Workshop* (2014).

[10] Sundararajan, M., Taly, A. & Yan, Q. *Axiomatic attribution for deep networks. Proceedings of the 34th International Conference on Machine Learning (ICML)*, Vol. 70, 3319–3328 (PMLR, 2017).

[11] Smilkov, D., Thorat, N., Kim, B., Viégas, F. & Wattenberg, M. *Smoothgrad: Removing noise by adding noise. ICML Workshop on Visualization for Deep Learning* (2017).

[12] Springenberg, J. T., Dosovitskiy, A., Brox, T. & Riedmiller, M. *Striving for simplicity: The all convolutional net. International Conference on Learning Representations (ICLR) Workshop* (2015).

[13] Selvaraju, R. R. *et al.* Grad-CAM: Visual explanations from deep networks via gradient-based localization. *International Journal of Computer Vision* **128**, 336–359 (2020). Originally arXiv:1610.02391 (2016).

[14] Chattopadhyay, A., Sarkar, A., Howlader, P. & Balasubramanian, V. N. *Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks. IEEE Winter Conference on Applications of Computer Vision (WACV)*, 839–847 (IEEE, 2018).

[15] Zeiler, M. D. & Fergus, R. *Visualizing and understanding convolutional networks. European Conference on Computer Vision (ECCV)*, 818–833 (2014).

[16] Fong, R. C. & Vedaldi, A. *Interpretable explanations of black boxes by meaningful perturbation. IEEE International Conference on Computer Vision (ICCV)*, 3429–3437 (2017).

[17] Petsiuk, V., Das, A. & Saenko, K. *RISE: Randomized input sampling for explanation of black-box models. British Machine Vision Conference (BMVC)* (2018).

[18] Ribeiro, M. T., Singh, S. & Guestrin, C. *"why should i trust you?": Explaining the predictions of any classifier. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1135–1144 (2016).

[19] Lundberg, S. M. & Lee, S.-I. *A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems (NeurIPS)* (2017).

[20] Such, F. P. *et al. An Atari model zoo for analyzing, visualizing, and comparing deep reinforcement learning agents. Proceedings of the 28th International Joint Conference on Artificial Intelligence (IJCAI)*, 3260–3267 (2019).

[21] Vig, J. *et al. Investigating gender bias in language models using causal mediation analysis. Advances in Neural Information Processing Systems (NeurIPS)* (2020).

[22] Meng, K., Bau, D., Andonian, A. & Belinkov, Y. *Locating and editing factual associations in GPT. Advances in Neural Information Processing Systems (NeurIPS)* (2022).

[23] Wang, K., Variengien, A., Conmy, A., Shlegeris, B. & Steinhardt, J. *Interpretability in the wild: A circuit for indirect object identification in GPT-2 small. International Conference on Learning Representations (ICLR)* (2023).

[24] Conmy, A., Mavor-Parker, A. N., Lynch, A., Heimersheim, S. & Garriga-Alonso, A. *Towards automated circuit discovery for mechanistic interpretability. Advances in Neural Information Processing Systems (NeurIPS)* (2023).

[25] Geiger, A., Lu, H., Icard, T. & Potts, C. *Causal abstractions of neural networks. Advances in Neural Information Processing Systems (NeurIPS)* (2021).

[26] Geiger, A., Wu, Z., Potts, C., Icard, T. & Goodman, N. D. *Finding alignments between interpretable causal variables and distributed neural representations. Proceedings of the Third Conference on Causal Learning and Reasoning (CLeaR)* (2024).

[27] Cunningham, H., Ewart, A., Riggs, L., Huben, R. & Sharkey, L. Sparse autoencoders find highly interpretable features in language models (2023). 2309.08600.

[28] Alain, G. & Bengio, Y. *Understanding intermediate layers using linear classifier probes. International Conference on Learning Representations (ICLR) Workshop* (2017).

[29] Belrose, N. *et al.* Eliciting latent predictions from transformers with the tuned lens (2023). 2303.08112.

[30] Nanda, N. Attribution patching: Activation patching at industrial scale. https://www.neelnanda.io/mechanistic-interpretability/attribution-patching (2023).

[31] Karvonen, A. *et al. SAEBench: A comprehensive benchmark for sparse autoencoders in language model interpretability. Proceedings of the 42nd International Conference on Machine Learning (ICML)*, Vol. 267 of *PMLR* (2025). Eight metrics over >200 open-sourced SAEs; finds gains on proxy metrics do not reliably translate to practical interpretability., 2503.09532.

[32] Jacovi, A. & Goldberg, Y. *Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness? Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 4198–4205 (Association for Computational Linguistics, 2020).

[33] Adebayo, J. *et al. Sanity checks for saliency maps. Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 31, 9505–9515 (2018).

[34] Paper 2 authors. Faithful-method demonstration: causal near-ceiling versus popular near-chance on the differentiable Atari VCS. Committed analysis record,

tools/xai_study/compare/out/faithful_demo.md; pure read over the committed cross-tradition leaderboard (compare/out/leaderboard.json) and the per-game §R records it aggregates. Generated by compare/faithful_demo.py. (2026). Internal data artifact. Position-regime pair activation_patching = 1.000 vs vanilla_saliency = 0.000 on all six core games (per-method gap 1.000). Accessed 2026-06-23.

[35] Paper 2 authors. Cross-tradition faithfulness–plausibility leaderboard for the differentiable Atari VCS interpretability audit (phases a–c). Committed analysis record, tools/xai_study/compare/out/leaderboard.json; aggregates the per-game §R records under tools/xai_study/phaseA_kording/out/ (2026). Internal data artifact; 257 per-game records aggregated into 31 method rows including the oracle positive control. Generated by compare/leaderboard.py. Accessed 2026-06-23.

[36] Yang, M. & Kim, B. BAM: Benchmarking attribution methods with relative feature importance (2019). Introduces the BAM benchmark: models trained with known relative feature importance plus paired-model/paired-input metrics for attribution methods. (Key yang2019bim retained for citation stability; the work is BAM, not BIM.), 1907.09701.

[37] Li, X. *et al.* *$\mathcal{M}^4$: A unified XAI benchmark for faithfulness evaluation of feature attribution methods across metrics, modalities and models.* *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, Vol. 36 (2023).

[38] Lindner, D. *et al.* *Tracr: Compiled transformers as a laboratory for interpretability.* *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 36 (2023). 2301.05062.

[39] Gupta, R., Arcuschin, I., Kwa, T. & Garriga-Alonso, A. *InterpBench: Semi-synthetic transformers for evaluating mechanistic interpretability techniques.* *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, Vol. 37 (2024). 2407.14494.

[40] Mueller, A. *et al.* *MIB: A mechanistic interpretability benchmark.* *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, Vol. 267 of *PMLR* (2025). Two tracks (circuit localization; causal-variable localization) over four tasks and five neural language models., 2504.13151.

[41] Bellemare, M. G., Naddaf, Y., Veness, J. & Bowling, M. The Arcade Learning Environment: An evaluation platform for general agents. *Journal of Artificial Intelligence Research* **47**, 253–279 (2013).

[42] Machado, M. C. *et al.* Revisiting the Arcade Learning Environment: Evaluation protocols and open problems for general agents. *Journal of Artificial Intelligence Research* **61**, 523–562 (2018).

[43] Dalton, S., Frosio, I. & Garland, M. *Accelerating reinforcement learning through GPU Atari emulation. Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33 (2020). ArXiv:1907.08467; code at https://github.com/NVlabs/cule.

[44] Delfosse, Q. *et al.* JAXAtari: A high-performance framework for reasoning agents. https://github.com/k4ntz/JAXAtari (2026). Software; accessed 2026-06-19.

[45] Barbiero, P. *et al.* Neural interpretable reasoning. *arXiv preprint arXiv:2502.11639* (2025).

[46] Marr, D. *Vision: A Computational Investigation into the Human Representation and Processing of Visual Information* (W. H. Freeman and Company, 1982).

[47] Lazebnik, Y. Can a biologist fix a radio? — or, what i learned while studying apoptosis. *Cancer Cell* **2**, 179–182 (2002).

[48] Bengio, Y., Léonard, N. & Courville, A. Estimating or propagating gradients through stochastic neurons for conditional computation (2013). 1308.3432.

[49] Jang, E., Gu, S. & Poole, B. *Categorical reparameterization with Gumbel-softmax. International Conference on Learning Representations (ICLR)* (2017).

[50] Jaderberg, M., Simonyan, K., Zisserman, A. & Kavukcuoglu, K. *Spatial transformer networks. Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 28 (2015).

[51] Abnar, S. & Zuidema, W. *Quantifying attention flow in transformers. Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 4190–4197 (2020).

[52] Chan, L. *et al.* Causal scrubbing: A method for rigorously testing interpretability hypotheses. AI Alignment Forum (2022).

[53] Shrikumar, A., Greenside, P. & Kundaje, A. *Learning important features through propagating activation differences. Proceedings of the 34th International Conference on Machine Learning (ICML)*, 3145–3153 (2017).

[54] Erion, G., Janizek, J. D., Sturmfels, P., Lundberg, S. M. & Lee, S.-I. Improving performance of deep learning models with axiomatic attribution priors and expected gradients. *Nature Machine Intelligence* **3**, 620–631 (2021).

[55] Štrumbelj, E. & Kononenko, I. Explaining prediction models and individual predictions with feature contributions. *Knowledge and Information Systems* **41**, 647–665 (2014).

[56] Fong, R., Patrick, M. & Vedaldi, A. *Understanding deep networks via extremal perturbations and smooth masks. Proceedings of the IEEE/CVF International*

Conference on Computer Vision (ICCV), 2950–2958 (2019).

[57] Olson, M. L., Khanna, R., Neal, L., Li, F. & Wong, W.-K. Counterfactual state explanations for reinforcement learning agents via generative deep learning. *Artificial Intelligence* **295**, 103455 (2021).

[58] Bastani, O., Pu, Y. & Solar-Lezama, A. *Verifiable reinforcement learning via policy extraction. Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 31, 2494–2504 (2018).

[59] Balduzzi, D. *et al. The shattered gradients problem: If resnets are the answer, then what is the question? Proceedings of the 34th International Conference on Machine Learning (ICML)*, Vol. 70 of *PMLR*, 342–350 (2017).

[60] Morcos, A. S., Barrett, D. G. T., Botvinick, M. & Rabinowitz, N. C. *On the importance of single directions for generalization. International Conference on Learning Representations (ICLR)* (2018). 1803.06959.

[61] Leavitt, M. L. & Morcos, A. S. *Towards falsifiable interpretability research. NeurIPS 2020 Workshop: ML Retrospectives, Surveys & Meta-Analyses (ML-RSA)* (2020). 2010.12016.

[62] Elhage, N. *et al.* Toy models of superposition (2022). Transformer Circuits Thread, https://transformer-circuits.pub/2022/toy_model/index.html, 2209.10652.

[63] Kindermans, P.-J. *et al.* in *The (un)reliability of saliency methods Explainable AI: Interpreting, Explaining and Visualizing Deep Learning*, Vol. 11700 of *Lecture Notes in Computer Science* 267–280 (Springer, 2019).

[64] Sturmfels, P., Lundberg, S. & Lee, S.-I. Visualizing the impact of feature attribution baselines. *Distill* **5** (2020). https://distill.pub/2020/attribution-baselines.

[65] Jain, S. & Wallace, B. C. *Attention is not explanation. Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, 3543–3556 (Association for Computational Linguistics, 2019).

[66] IEEE. IEEE standard glossary of software engineering terminology (ieee std 610.12-1990). IEEE, New York (1990). Defines “software” as computer programs, procedures, and possibly associated documentation and data.

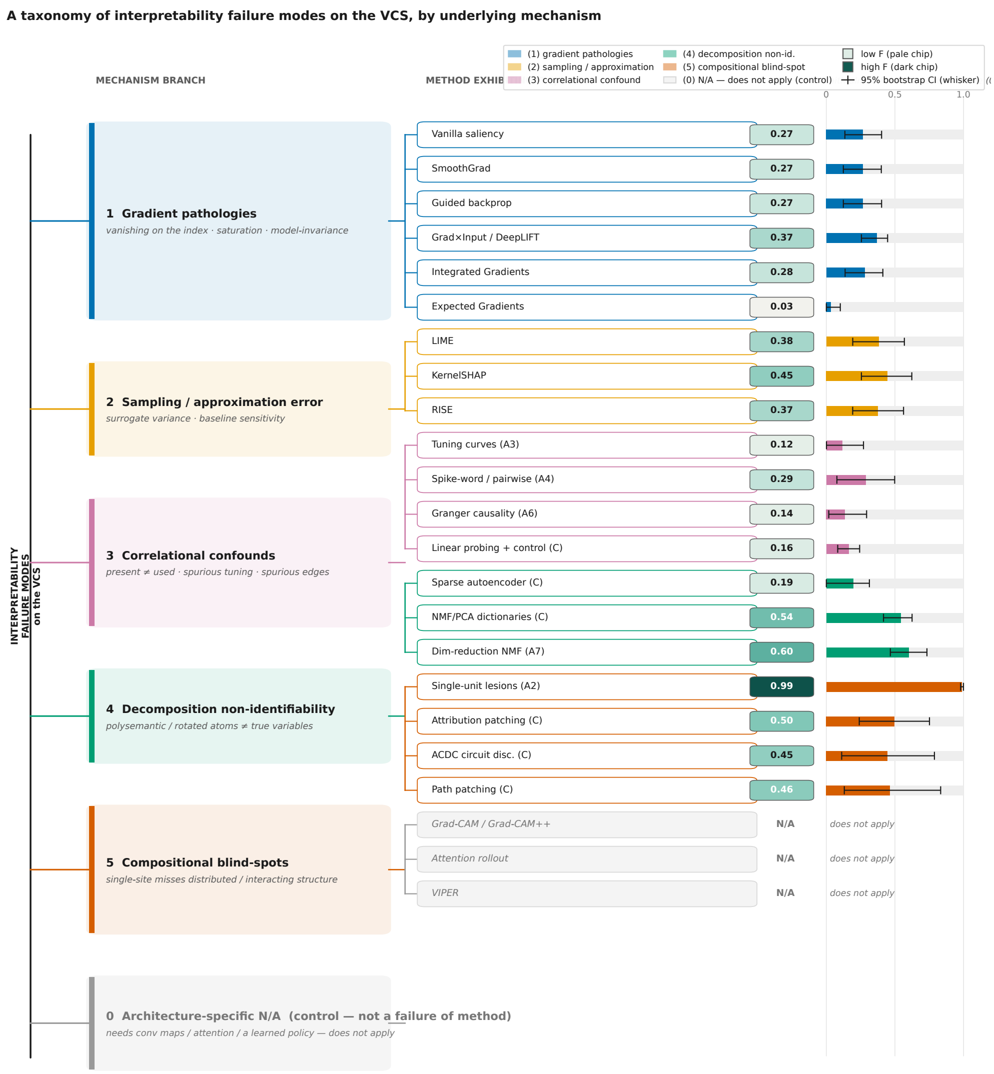

Fig. 5 A taxonomy of interpretability failure on the VCS, organised by mechanism. Each leaf is a method exhibiting the branch's failure mode, annotated with its *family-mean* faithfulness $F \pm 95\%$ bootstrap-over-games CI against the intervention oracle (Section 3.2; $0 =$ chance, $1 =$ oracle ceiling), read from leaderboard_ci.csv; the per-game exemplars quoted below carry their own "on ⟨game⟩" label. The lower branches confirm the expected failures: the gradient family is provably zero on the discrete position/index regime, and the architecture-specific methods (Grad-CAM, attention, VIPER) do not apply because the VCS lacks the neural substrate they require, a property of the subject, not a method failure. The higher-order branches are the *semantic gap*: present-but-unused concepts (linear probing [8] flags a present-but-unused cell the oracle says is inert), named-but-inert factorisations (the SAE [27] matches every variable at fraction 1.0 yet scores $F = 0.041$ on Pong), and graph-correct-but- incomplete circuits (ACDC [24] at $F = M = 1.0$, $S = 0.44$ on Breakout). The neural-network parallels shown alongside each branch are documented analogues [59–65], evidence transfer rather than proof. Every annotated value reads from a committed record.

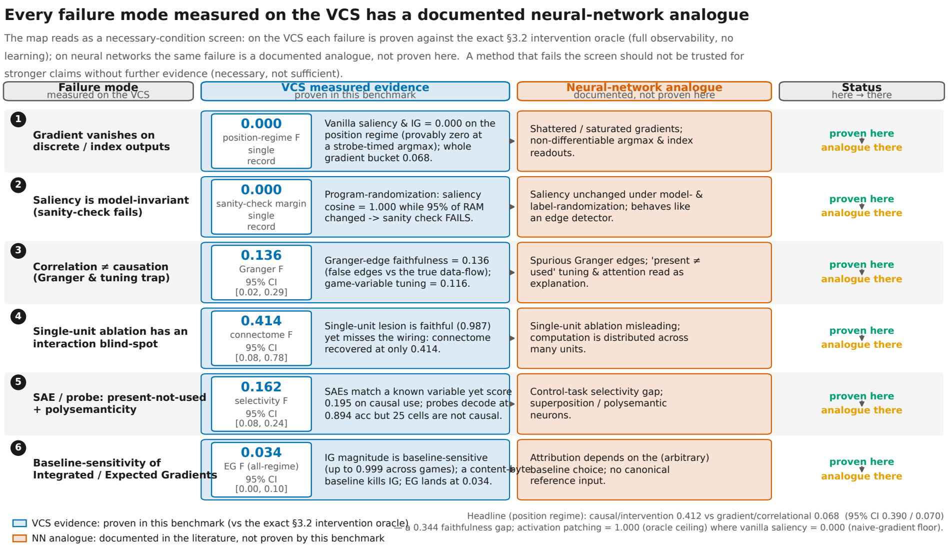

Fig. 6 The VCS↔neural-network failure-mode map. Each row pairs a failure mode we measured on the VCS against the oracle (§3.2) — left, blue — with the neural-network behavior it is a documented analogue of — right, vermillion. On the VCS the failure is provable because we hold the exact ground truth; the neural-network column is a documented analogue, not proven by this benchmark, and reads the attribution and probing literature ([10, 33, 59, 64] on gradients; [8, 60, 61, 65] on use vs. presence; [6, 62, 63] on probing, superposition, and reliability). The takeaway is one line: every VCS failure has a named counterpart in how interpretability methods are reported to fail on networks, so a property provable here is one to suspect there. Per-row VCS scores, their committed records, and the full map (`fig5_representativeness_map_full.pdf`) are in the Supplementary Information.

Supplementary Information

This Supplementary Information records what the main paper summarizes. We give the per-method protocols (Sec. S1), the leaderboard schema and its glossary (Sec. S2), the coverage and applicability of every method across games and output regimes (Sec. S3), the mapping from each headline claim to its evidence (Sec. S4), the provenance of every headline number down to the record file and command that produced it (Sec. S5), a plain statement of what the paper does not claim (Sec. S6), and the plausibility-proxy rubric together with its sensitivity analysis (Sec. S7). The two full-resolution diagnostic figures whose simplified forms appear in the main text are reproduced here at full density (Figs. S1 and S2). The correctness triad $F \wedge S \wedge M$ used throughout is the one defined in the main Methods (Section 3.4); we reference it here rather than restating it. The benchmark scores 30 interpretability methods together with the intervention oracle (Section 3.2) as a positive control, for 31 leaderboard rows in total.

S1 Per-method protocols

This section records, for each method under test, the exact objects it consumes and produces, the reference it is scored against, its sampling or optimization budget, the metric that turns its output into a faithfulness score, and the committed record file that holds the per-game results. All methods share one evaluation contract, which we state once and then specialize per method. Every method receives the same bit-exact VCS trajectory on one of the six core games (`breakout`, `ms_pacman`, `pong`, `qbert`, `seaquest`, `space_invaders`); the candidate-cause set is the machine's own state, that is the 128 RAM bytes together with the CPU registers and the TIA and RIOT registers exposed by the substrate; the reference is the intervention oracle of Section 3.2, evaluated by exact re-runs of the emulator under $\mathrm{do}(u)$; and a method that cannot be applied to a given output regime is recorded as not-applicable rather than scored as zero, so that an empty result never masquerades as a wrong one. Faithfulness is the F axis of the main-text triad (Section 3.4); we reference that definition and do not restate it. Records live under `tools/xai_study/<phase>/out/`; the aggregate they feed is `tools/xai_study/compare/out/leaderboard.json`.

The neuroscience battery (Phase A) treats the running machine as a brain and asks how far each classical analysis recovers the known mechanism. Each method's baseline is the exact data-flow graph or causal role that the substrate makes available, its budget is the six-game core unless noted, and its scoring metric is named in Table S1.

The attribution battery (Phase B) runs the everyday XAI toolbox over the same trajectories. These methods produce a real-valued saliency or attribution over the candidate-cause set, which we score by Pearson correlation against the oracle attribution; the gradient-based members additionally inherit the gradient-vanishing failure of the discrete position regime (Prop. 1). The sampling budgets that matter are the Monte-Carlo or perturbation counts of the sampling estimators, recorded per method in Table S2; the seed-dispersion of the stochastic members is carried in `leaderboard_ci.csv` (`seed_sd`, `seed_sd_source`) and reproduced in Sec. S5.

Table S1 Phase A protocols (neuroscience battery). One row per method: the input it reads, the reference it is scored against, the sampling or intervention budget, the scoring metric, and the committed record. All eight run on the six core games; faithfulness is the F axis of Section 3.4. The whole-state recording (A8) is included as a descriptive upper bound on coverage, not as a faithful explanation, and is read for its minimality M rather than its trivial F.

Method	Input / candidate causes	Reference	Scoring metric	Record
A1 connectomics	RAM/register activity over trajectory	data-flow graph	data-flow graph F_1 vs. oracle	A1
A2 lesions	single-cell knockouts via do(u)	true causal role	lesion-importance ρ vs. role	A2
A3 tuning curves	per-cell response vs. variable	variable coupling	$1-$ spurious-tuning rate	A3
A4 spike-word / pairwise	pairwise cell correlations	true coupling	corr. ρ vs. coupling	A4
A5 field potentials	pooled-activity spectra	true causal share	$1-$ clock-explained var.	A5
A6 Granger	directed lag structure	true data-flow	$1-$ false-edge rate	A6
A7 dim. reduction	NMF/PCA over activity	known variables	NMF matched-component frac.	A7
A8 whole-state	full state record	oracle decomp.	minimality M vs. oracle	A8

Records: `tools/xai_study/phaseA_kording/out/A{1..8}_*`; all eight run on the six core games.

The mechanistic battery (Phase C) runs the methods built to recover circuits, scoring each against the true data-flow or the exact recovered-minus-exact deviation. The causal members of this phase are the ones that reach the oracle ceiling, and we list their exact scoring metric so that the ceiling is not mistaken for a tautology: each is scored against the substrate's known routine, not against another method. Two members (NMF/PCA dictionaries and the sparse autoencoder) are dimensionality-reduction methods carried here for comparison, and the sparse autoencoder is the one whose two aggregations differ, which we reconcile in Sec. S5.

The intervention oracle is the positive control, not a method under test. It is the exact do(u) ground truth of Section 3.2, evaluated by re-running the bit-exact machine, and it attains $F = S = M = 1.0$ by construction; its record triple is the intervention oracle, the content-path gradient oracle, and their cross-check (`tools/xai_study/ground_truth/out/`). It anchors the ceiling against which every other row is read, which is why it occupies the thirty-first leaderboard row.

S2 Benchmark schema and glossary

The leaderboard is built from a flat record schema, and stating it removes the ambiguity about what a single number on the benchmark counts. Each method is run per game and writes a per-game record under its phase's `out/` tree; the leaderboard aggregates those records into one row per method by averaging over games. The schema

Table S2 Phase B protocols (attribution battery). Each method emits a saliency over the candidate-cause set and is scored by Pearson correlation against the oracle attribution (Section 3.2). The reference and the six-game budget are shared; the budget column records the sampling or perturbation count that drives each estimator, and the seed column flags which methods carry an independent-seed dispersion in `leaderboard_ci.csv`.

Method	Tradition	Budget / hyperparameters	Seed sd
`vanilla_saliency`	gradient	single backward pass	—
smoothgrad	gradient	noise-averaged gradient; σ-robustness committed	0.0
`guided_backprop`	gradient	single guided pass	—
`gradxinput_deeplift`	gradient	DeepLIFT reference-difference	—
`integrated_gradients`	gradient	path integral, baseline = zero state	—
`expected_gradients`	gradient	expectation over baselines (refits)	yes
kernelshap	gradient	Monte-Carlo coalition sampling ($\propto 1/\sqrt{N}$)	0.017
lime	gradient	local linear fit, perturbation refits	0.029
rise	gradient	random-mask Monte-Carlo ($\propto 1/\sqrt{N}$)	0.051
occlusion	intervention	exhaustive single-cell masking	—
`extremal_perturbation`	intervention	optimized mask, area-constrained	—
`on_distribution_counterfactual`	intervention	on-manifold do(u) resampling	—

Records: `tools/xai_study/phaseB_attribution/out/`; all twelve run on the six core games and are scored by Pearson correlation against the oracle (Section 3.2). Seed sd is the independent-seed dispersion from `leaderboard_ci.csv`.

columns of a leaderboard row are given in Table S4; the aggregation rule is a mean over the n_{games} games (with the per-game records first reduced within a game when a method emits several output records), and the faithfulness, sufficiency, and minimality columns report the F, S, and M axes defined in Section 3.4. We do not re-derive those definitions here.

The vocabulary that the schema implies is worth fixing precisely, because the review asked for a single unambiguous method count. We distinguish a *method* (one interpretability technique, e.g. `vanilla_saliency`), a *method row* (its single aggregated line on the leaderboard), a *per-game record* (one method scored on one game, written under a phase `out/` tree), an *output record* (a per-game record for one output regime, several of which a method may emit on one game), a *family mean* (the average over the members of a tradition family, e.g. causal/intervention), the *oracle positive control* (the exact do(u) ground truth of Section 3.2, scored as a method to anchor the ceiling), and a *not-applicable method* (one that does not apply to a given output regime and is recorded as N/A rather than as a zero). With this vocabulary the count is unambiguous. The benchmark scores 30 interpretability methods together with the intervention oracle as a positive control, for 31 leaderboard rows in total (`leaderboard.json.n_methods` = 31), aggregating 257 per-game records

Table S3 Phase C protocols (mechanistic battery). Each method is scored against the substrate's known circuit or the exact recovered-minus-exact deviation (Section 3.2), not against another method, which is what keeps the oracle ceiling non-circular. The probing and dictionary-learning members are included for comparison with the causal members.

Method	Tradition	Scoring metric
`activation_patching`	causal	max $\lvert$recovered − exact$\rvert$ vs. oracle
`causal_scrubbing`	causal	scrubbing-preserved performance
`interchange_interventions_das`	causal	aligned interchange accuracy
`logit_tuned_lens`	causal	logit-lens fidelity to true intermediate
`path_patching`	causal	path-circuit F_1 vs. true routine
ACDC	causal	discovered-circuit best F_1 vs. data-flow
`attribution_patching`	gradient	corr. of approximation vs. exact patch
`linear_probing_control_tasks`	probing	selectivity vs. control task
`nmf_pca_dictionaries`	dim. red.	matched-component fraction vs. vars
`sparse_autoencoder`	dim. red.	feature-variable matched fraction

Records: `tools/xai_study/phaseC_mechanistic/out/`; all ten run on the six core games.

(`n_per_game_records_aggregated` = 257): 8 methods in Phase A, 12 in Phase B, 10 in Phase C, and the one oracle row. This count is the canonical one used in the abstract, the figures, and throughout this Supplement.

S3 Coverage and applicability

The benchmark scores methods on a six-game core drawn from a larger bit-exact substrate, and the coverage table makes the scope of that choice explicit. Paper 1 validates the differentiable VCS ports bit-for-bit against the `xitari` reference on all 64 Arcade Learning Environment games, within a 30-frame RAM and 60-frame screen window; the interpretability study reported here runs on the six games for which all three tiers of ground truth (Section 3.3) are available and the oracle is exact, listed in Table S5. The remaining 58 games are bit-exact substrate but are not scored here, either because they fall outside the validated horizon for the analyses we run or because the tiered labels they would need are not yet produced; they are candidates for the larger sweep rather than excluded for any methodological reason. The horizon column records the validated window, and the tier column records which of the three ground-truth tiers (T1 exact intervention, T2 data-flow, T3 externally-supplied semantic labels) is available.

Whether a method applies at all depends on what the method needs from its target, and the applicability matrix records that for every method family. The Phase A neuroscience analyses read the raw machine state and need neither a learned policy nor gradients; the Phase B attribution methods split into gradient members, which need a differentiable forward pass, and intervention members, which need only the ability to set state; the Phase C mechanistic methods need access to internal activations and, for the causal members, the ability to intervene. Table S6 records, per family, whether

Table S4 Leaderboard row schema. The columns of one row in `compare/out/leaderboard.json` (`rows[]`) and its uncertainty companion `leaderboard_ci.csv`. The metric column names the per-method scoring function from Sec. S1; F, S, and M are the triad axes of Section 3.4. Aggregation is a mean over n_{games}, with bootstrap confidence intervals over games (and, where present, over records).

Column	Meaning
`phase`	phaseA_kording, phaseB_attribution, phaseC_mechanistic, or ground_truth
`method`	method identifier (the row key)
`tradition`	intervention, causal, gradient, correlational, `dim_reduction`, probing, descriptive, oracle
`n_games`	number of games the method was scored on (6 for the core set; 1 for the oracle)
`n_records`	number of per-game output records aggregated into the row
`faithfulness` (F)	faithfulness vs. the intervention oracle (Section 3.2), $[0, 1]$, higher better
`faithfulness_ci95`	95% bootstrap CI half-width over games
`faithfulness_position_regime`	F restricted to the discrete sprite-position/index regime
`faithfulness_content_regime`	F restricted to the continuous content regime
`triad_S` / `triad_M`	sufficiency / minimality axes (Section 3.4), where applicable
`plausibility_proxy`	documented method-tradition proxy, $[0, 1]$ — not a human-subjects measurement
`is_positive_control`	1 for the oracle row, 0 otherwise
`metric_names`	the per-method scoring function (Sec. S1)
`commits`	the commit(s) at which the per-game records were produced

Table S5 Core-game coverage. The six core games carry exact tiered ground truth and the full method battery; faithfulness is scored within the Paper 1 bit-exact window. The last two columns give the methods run and the per-game output records contributed. The wider 64-game substrate is bit-exact (Paper 1) but is reserved for the larger sweep; T3 labels are external (Section 3.3), which is recorded so the benchmark does not appear to certify them.

Game	Paper-1 status	Horizon	Tiers	Methods run	Records
breakout	bit-exact	30/60 fr	T1,T2,T3	30 + oracle	per-game
ms_pacman	bit-exact	30/60 fr	T1,T2,T3	30 + oracle	per-game
pong	bit-exact	30/60 fr	T1,T2,T3	30 + oracle	per-game
qbert	bit-exact	30/60 fr	T1,T2,T3	30 + oracle	per-game
seaquest	bit-exact	30/60 fr	T1,T2,T3	30 + oracle	per-game
space_invaders	bit-exact	30/60 fr	T1,T2,T3	30 + oracle	per-game
58 further games	bit-exact (P1)	30/60 fr	T1,T2 partial	reserved	—

the method operates on the raw VCS, whether it requires a learned policy, gradients, internal layers or attention, or external labels, whether it uses oracle-like interventions, and whether it receives a supplied hypothesis. The pattern the table exposes is the one the results confirm: the families that earn faithfulness are exactly those that can intervene, and the families that collapse on the position regime are exactly the gradient and correlational ones.

Table S6 Per-family applicability. What each method family requires of its target. "Intervenes" marks oracle-like do(u) access; "hypothesis" marks methods that receive a supplied circuit hypothesis to test. The raw-VCS column distinguishes analyses that read machine state directly from those that need a differentiable or layered model. A blank means the requirement does not apply.

Family	raw VCS	policy	gradients	layers/attn	intervenes	hypothesis
A neuroscience	yes	no	no	no	A2 only	no
B gradient	yes	no	yes	no	no	no
B intervention	yes	no	no	no	yes	no
C causal	yes	no	no	yes	yes	yes
C gradient (attr. patch)	yes	no	yes	yes	approx.	yes
C probing	yes	no	no	yes	no	no
C / A dim. reduction	yes	no	no	yes	no	no
oracle (control)	yes	no	no	yes	exact	—

S4 Claims and evidence

Every headline claim in the paper rests on a specific artifact, and the claims-and-evidence table makes that dependency checkable. We list each claim, the figure or table that carries it, the games and methods it draws on, the metric, the artifact it traces to, and the limitation that bounds it. The faithfulness, sufficiency, and minimality references are the triad axes of Section 3.4, and the numbers themselves are sourced in Sec. S5; the table below names which evidence supports which claim, not the values.

S5 Number provenance

Every headline number traces to a committed record, the script that produced it, and the command that re-derives it, and this section is the in-paper copy of that traceability table. It mirrors Section 6.5 of the reproducibility document (`xai_2_interpretability/REPRODUCIBILITY.md`); the values are the measured, committed ones at the current revision, and the confidence intervals are read from `tools/xai_study/compare/out/leaderboard_ci.csv` rather than recomputed here. Each record additionally carries `seed`, `where`, `commit`, and `timestamp`, so the provenance of every value is self-documenting.

One number requires a reconciliation, and we record it rather than hide it. The sparse autoencoder is the only method whose two aggregations disagree: the over-games mean is $F = 0.1404$ with a 95% bootstrap CI of [0.0, 0.3139], while the over-records mean is 0.1947 with CI [0.0058, 0.4804], both read from `leaderboard_ci.csv`. The figures and the main-text leaderboard plot the over-records value 0.1947, which lies inside the over-games interval; we treat the over-games mean as the canonical row value for the aggregation rule of Sec. S2 and report the over-records figure alongside it, so that text, table, and figure agree on which aggregation each cites. No headline claim depends on which of the two is read, since the sparse autoencoder sits near the floor under both.

Table S7 Claims and their evidence. Each headline claim maps to the figure or table that carries it, the games and methods involved, the metric, the committed artifact, and the limitation that bounds the claim. The artifact column points at the record that Sec. S5 traces to a value; the limitation column states the scope so the claim is not read wider than the evidence.

Claim	Evidence	Methods / games	Metric	Limitation
Causal/intervention methods beat gradient/correlational on faithfulness across all regimes	Fig. 4; leaderboard	25 methods, 6 games	F mean (gap 0.387)	not robust: gap collapses to 0.123 $[-0.008, 0.256]$ excl. oracle-like and 0.045 $[-0.087, 0.153]$ content-only (CIs cross zero); only the position-regime gap 0.3435 is robust
The gap holds with the same sign on the discrete position regime	Fig. 4; `faithful_demo`	13 rows, 6 games	F position (gap 0.344)	position regime only
A faithful causal method reaches the oracle ceiling where the default tool collapses	`faithful_demo`; Fig. 4	act. patching vs. vanilla saliency, 6 games	F 1.0 vs. 0.0 (position regime; all-regimes saliency F 0.267, plotted in Fig. 4)	one decisive pair, position regime only
The naive gradient of a discrete index output is provably zero	Prop. 1 (Section 3.2); `oracle_grad`	all gradient methods	grad@index $= 0$	index path only
High plausibility coexists with low faithfulness (the danger zone)	Fig. 4; `faithful_demo`	vanilla saliency	proxy 0.9, F 0.0 (position regime; all-regimes F 0.267, plotted in Fig. 4)	proxy, not measured plausibility
Faithful attribution recovers the wiring but not the computation's semantics	Fig. 5 (Section S3); discussion	Phase B vs. C	F vs. S, M	necessary-condition stress test
The failure modes have documented NN analogues	Fig. 6 (Fig. S1)	taxonomy	—	analogues, not proof

S6 What this paper does not claim

The benchmark is sharp precisely because its scope is narrow, and stating the boundaries keeps the result from being read wider than the evidence carries it. We do not claim to recover a program's semantics from scratch. The study measures whether an explanation names the true causes and regenerates the behavior on a machine whose answer is already known; it does not show that any method discovers the meaning of a routine without that ground truth in hand, and the residual that faithfulness alone leaves behind, the gap between the wiring and the computation, is the point rather than a result to be explained away.

We do not evaluate learned agents. The subject throughout is the bit-exact VCS itself, and the candidate causes are its registers, RAM cells, and program variables, not the weights of a trained policy. The neural-network parallels drawn in the discussion and in Fig. S1 are documented analogues from the literature, not claims proven on this benchmark; we label them as such so that a reader does not take a mapped failure mode for a demonstrated one.

We do not prove that neural-network interpretability is impossible. The VCS gap is a necessary-condition stress test: a method that fails to recover the wiring of a fully known machine cannot be trusted to recover it on an unknown one, but a method that succeeds here is not thereby certified on networks. The “floor” framing is an evidence-transfer argument, not a proven lower bound on neural-network interpretability.

We do not measure human plausibility. The plausibility axis is a documented method-tradition proxy, computed from a fixed rubric rather than collected from raters, and we report it as a proxy throughout; its job is to expose the danger zone where a convincing explanation is unfaithful, not to stand in for a human-subjects study. We also do not redistribute the ROMs: the substrate is validated against an external reference, the records are reproducible from committed artifacts without the ROMs, and the ROM provenance is given as hashes so a third party supplies their own legally obtained copies.

S7 Plausibility-proxy rubric and its sensitivity

The plausibility axis of the leaderboard is a documented, tradition-level proxy, not a human-subjects measurement, and this section records how each value was assigned and shows that the headline contrast does not depend on the particular numbers chosen. The Methods (Section 3.4) name two reporting axes: faithfulness against the intervention oracle of Section 3.2, which is measured on the machine, and a plausibility proxy, which stands in for how convincing a tradition’s explanations are usually taken to be. We assign the proxy once per method tradition, before any faithfulness score on our machine is computed, so it cannot be tuned to the result. Its only job is to expose the danger zone of explanations that look convincing yet are unfaithful, and to that end it must be coarse, fixed in advance, and sourced rather than precise.

The rubric assigns one proxy per tradition on a fixed five-step scale, scoring two documented properties of each tradition’s typical output and taking their average. The first property is whether the explanation produces a dense, picture-like artifact that a reader sees as an account of the computation, since heatmaps and saliency-style maps over the input are read as convincing by construction; this is the perceptual-immediacy axis. The second is whether the field has historically treated the tradition’s output as a self-evident explanation rather than a hypothesis to be tested, which is the adoption-as-explanation axis. A tradition that scores high on both, such as gradient saliency, receives a high proxy; a tradition whose output is an austere intervention table or a circuit edge list, which is rarely mistaken for a finished story, receives a

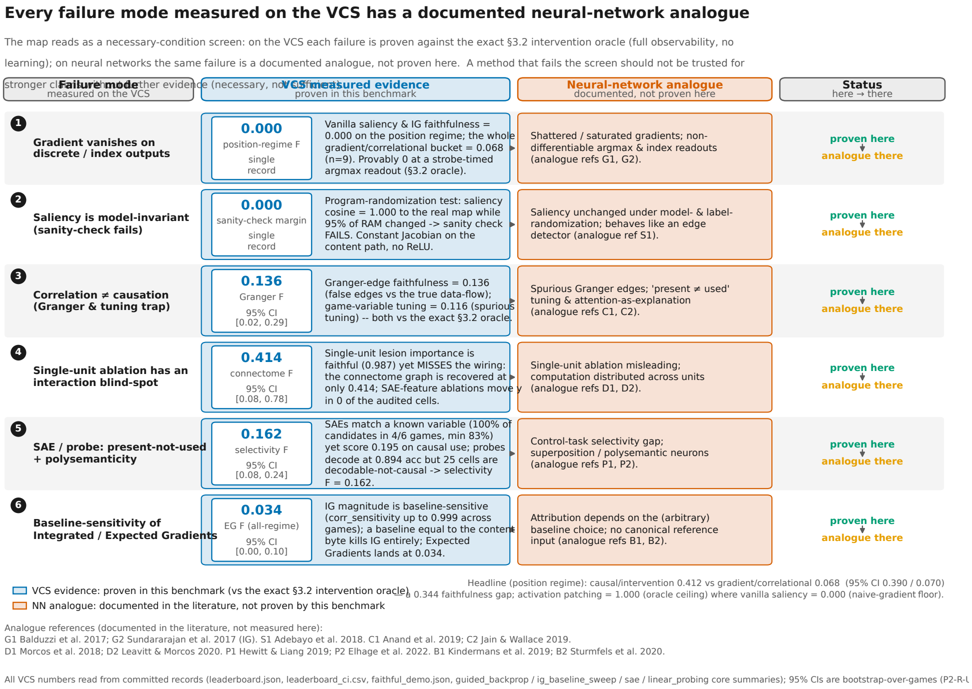

Fig. S1 Full VCS-to-NN representativeness map. The full-density version of the main-text map (Fig. 6): each VCS failure mode is placed against its documented neural-network analogue from the literature. The neural-network column records documented analogues, not parallels proven by this benchmark; the oracle is the intervention oracle of Section 3.2, and the underlying per-method values trace to `compare/out/leaderboard.json` via Sec. S5. Note that this full map is the figure the simplified main-text version condenses.

low one. The scale is deliberately blunt, $\{0.3, 0.5, 0.55, 0.75, 0.8, 0.85, 0.9\}$, because a proxy meant to expose a danger zone gains nothing from false precision and would invite over-reading if it were finer. Every value in Table S9 traces to the `plausibility_proxy` field of the per-method rows in `compare/out/leaderboard.json`; the table only groups those rows by tradition and records the rationale.

The contrast the paper draws is between this proxy and measured faithfulness, and on the committed records the two move in opposite directions. Across the thirty methods under test the Spearman rank correlation between faithfulness and the proxy is -0.64, a strong negative association: the traditions the field finds most convincing are, on this machine, among the least faithful. Fourteen methods fall in the danger zone, defined here as a proxy of at least 0.70 together with a faithfulness below 0.40, and they include the entire gradient battery, the correlational neuroscience methods, linear probing, and the sparse autoencoder. The per-method extreme makes the point sharply: vanilla saliency carries the highest proxy, 0.9, yet scores faithfulness 0.0 on the position regime, while activation patching carries a lower proxy, 0.5, and scores 1.0; the less faithful method is the more plausible one. These figures are produced by `tools/xai_study/compare/plausibility_sensitivity.py` from the leaderboard and recorded in `compare/out/plausibility_sensitivity.csv`.

A proxy fixed by a coarse rubric invites the question of whether the contrast is an artifact of the particular numbers, and the answer is that it is not. We perturb the proxy under four deterministic schemes and recompute the relationship each time; the analysis uses fixed, index-keyed offsets rather than random draws, so it re-derives bit-for-bit. The first scheme adds a fixed per-method jitter of magnitude $\sigma \in \{0.05, 0.1, 0.2\}$, alternating in sign by row. The second compresses every proxy toward the common mean by a fraction σ, which preserves the rank order of traditions. The third uses the same index-keyed offsets at $\sigma \in \{0.1, 0.2\}$, large enough to swap neighbouring traditions' ranks. The fourth drops one whole tradition at a time and recomputes on the remainder. Across all fourteen perturbations the rank correlation between faithfulness and proxy stays negative, ranging from -0.79 to -0.40 and never approaching zero; the danger zone stays populated, between six and fifteen methods; and the per-method anchor sign holds in every scheme in which both anchor methods are present, the more plausible saliency remaining the less faithful. The worst case is the leave-the-gradient-tradition-out run, which removes the most plausible and least faithful block at once and still leaves six danger-zone methods and a rank correlation of -0.79. Table S10 reports the full sweep. The headline contrast, that plausibility and faithfulness are not the same axis and that high plausibility is no guarantee of faithfulness, survives every alternative proxy assignment we tried.

Table S8 Headline-number provenance. Each number → its committed record and JSON key → the generating or aggregating script → the command that reproduces it. Values are the measured, committed numbers at this revision (consistent with REPRODUCIBILITY Section 6.5); CIs are read from `leaderboard_ci.csv`. Paths are relative to `tools/xai_study/`.

Number	Value	Record (JSON key)	Command
Faithfulness gap, position regime (causal − gradient)	0.3435	`faithful_demo`: `value`; `leaderboard`: `...position_regime.gap`	`faithful_demo.py`
Causal/interv. mean F, position regime	0.4118 ±0.39, n=4	`leaderboard`: `...position_regime.causal_..._mean`	`leaderboard.py`
Gradient/corr. mean F, position regime	0.0683 ±0.07, n=9	`leaderboard`: `...position_regime.gradient_..._mean`	`leaderboard.py`
Causal/interv. mean F, all regimes	0.6811 n=11	`leaderboard_ci.csv`: `family_causal_intervention` mean	`uncertainty.py`
Gradient/corr. mean F, all regimes	0.2938 n=14	`leaderboard_ci.csv`: `family_gradient_correlational` mean	`uncertainty.py`
Faithfulness gap, all regimes	0.3873	`leaderboard_ci.csv`: `all_regime_gap` mean	`uncertainty.py`
Decisive pair, position regime (act. patch vs. vanilla sal.)	1.000 vs. 0.000	`faithful_demo`: `pair_contrast`	`faithful_demo.py`
Plausibility proxy, decisive pair	0.5 vs. 0.9	`faithful_demo`: `...plausibility_proxy`	`faithful_demo.py`
ACDC sufficiency S	0.3247	`leaderboard_ci.csv`: ACDC `S_mean`	`uncertainty.py`
Whole-state minimality M	0.0703	`leaderboard_ci.csv`: A8 `M_mean`	`uncertainty.py`
Methods scored (rows)	31	`leaderboard`: `n_methods`	`leaderboard.py`
Per-game records aggregated	257	`leaderboard`: `n_per_game_records_aggregated`	`leaderboard.py`
Naive gradient, discrete index output	index 0; content 1.0	`oracle_grad_pong.json`: `value`	`oracle_grad.jl`
Exact oracle max $\lvert\Delta\text{score}\rvert$ (Pong)	17.0	`oracle_pong_score.json`: `value`	`oracle_intervene.jl`

Scripts live under `tools/xai_study/compare/` (Python) and `tools/xai_study/ground_truth/` (Julia); records under the corresponding `out/` trees. JSON keys are abbreviated with "`...`" where nested; the full paths are in REPRODUCIBILITY Section 6.5.

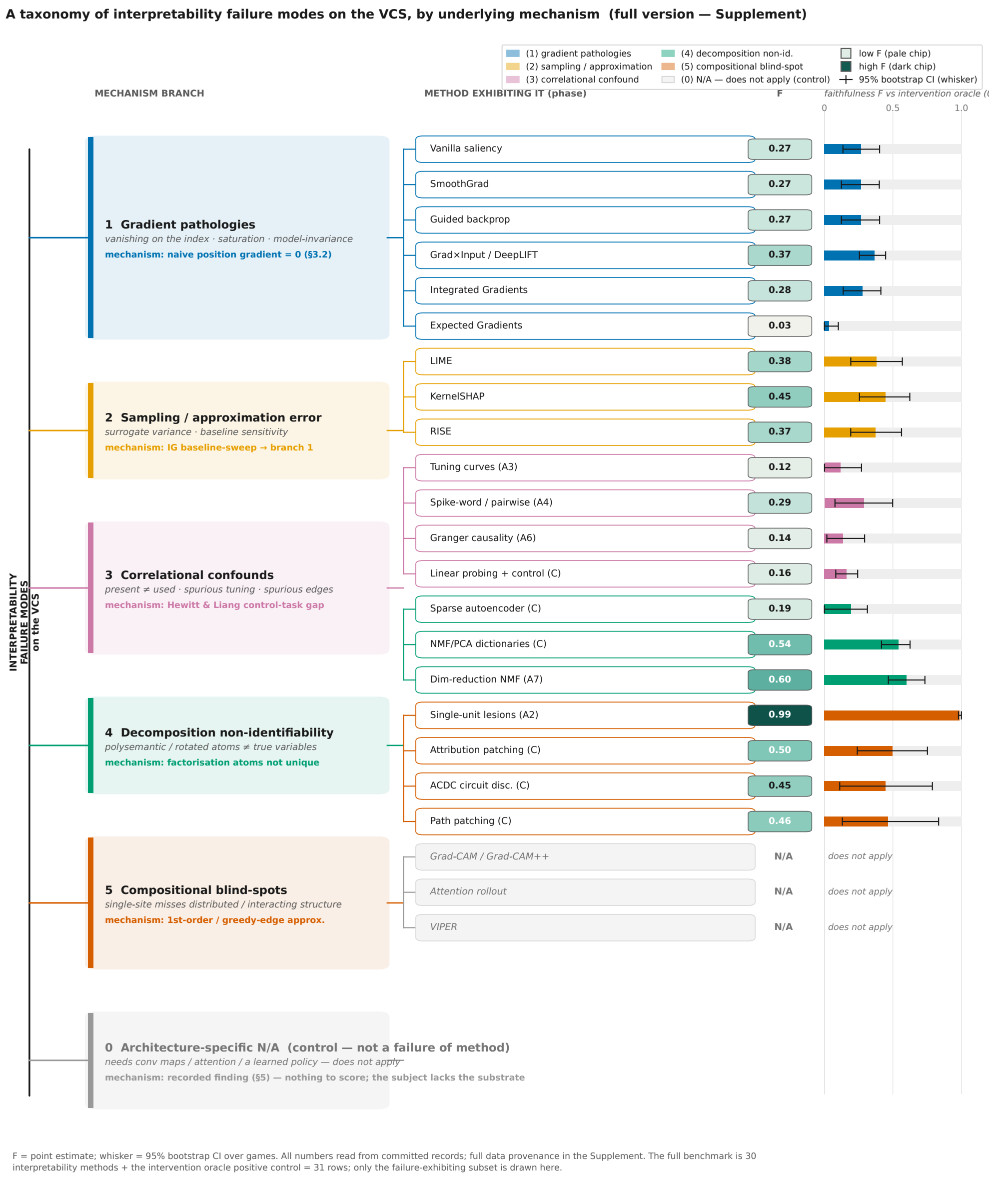

Fig. S2 Full failure taxonomy. The full-density version of the main-text taxonomy (Fig. 5): the families of interpretability failure observed on the machine, each annotated with its faithfulness against the intervention oracle (Section 3.2), reported as F with its 95% bootstrap confidence interval. The leaf values trace to `compare/out/leaderboard.json` and `leaderboard_ci.csv` via Sec. S5. Note that the gradient and correlational branches collapse on the discrete position regime, where the naive gradient is provably zero (Prop. 1).

Table S9 Plausibility-proxy rubric. The proxy assigned to each method tradition, with the documented rationale. Values are the `plausibility_proxy` field of the leaderboard rows (`compare/out/leaderboard.json`); they are tradition-level and fixed before any faithfulness score is computed, so the proxy cannot be tuned to the measured faithfulness. Higher means an explanation that the field tends to find more convincing, not one that is more faithful. The oracle (Section 3.2) is the positive control and is not a method under test.

Tradition	Proxy	Rationale (documented basis)
gradient	0.90	Dense input-space heatmaps read as a visual account of the decision; the most widely adopted XAI outputs, routinely shown as if self-explanatory.
probing	0.85	A trained decoder that "finds" a concept reads as evidence the concept is used; high adoption, though decodability is not causal use.
correlational	0.80	Tuning curves and coupling spectra match the neuroscience idiom of a convincing single-unit account.
dim. reduction	0.75	Low-dimensional factors and dictionary atoms look like a clean summary of the representation.
intervention	0.55	Occlusion and counterfactual outputs are quantitative tables, less immediately picture-like, treated as tests rather than stories.
causal	0.50	Circuit edge lists and patching results are austere artifacts rarely mistaken for a finished narrative.
descriptive	0.30	Whole-state recording is an explicit non-explanation: it withholds nothing and reads as a dump, not an account.
oracle (control)	1.00	The Section 3.2 ground truth by construction; positive control, not a method under test.

Table S10 Proxy-sensitivity sweep. The faithfulness-versus-proxy relationship under each perturbation, from `compare/out/plausibility_sensitivity.csv`. "ρ" is the Spearman rank correlation between faithfulness and proxy over the methods present; "danger" is the count of high-proxy (≥ 0.70), low-faithfulness (< 0.40) methods; "anchor" is 1 when the more plausible saliency is the less faithful of the saliency/activation-patching pair (blank where a leave-one-out run removes an anchor). The baseline rubric is the first row. Across every scheme the correlation stays negative and the danger zone stays populated.

Scheme	Param	n	ρ	Danger	Anchor
baseline	—	30	−0.64	14	1
additive jitter	$\sigma = 0.05$	30	−0.63	14	1
additive jitter	$\sigma = 0.10$	30	−0.61	14	1
additive jitter	$\sigma = 0.20$	30	−0.55	15	1
rank-preserving	$\sigma = 0.10$	30	−0.64	14	1
rank-preserving	$\sigma = 0.20$	30	−0.64	14	1
rank-jittering	$\sigma = 0.10$	30	−0.61	14	1
rank-jittering	$\sigma = 0.20$	30	−0.55	15	1
leave-out causal	—	24	−0.40	14	
leave-out correlational	—	26	−0.67	10	1
leave-out descriptive	—	29	−0.60	14	1
leave-out dim. red.	—	27	−0.64	13	1
leave-out gradient	—	20	−0.79	6	
leave-out intervention	—	25	−0.61	14	1
leave-out probing	—	29	−0.65	13	1